TOPOLOGICAL SEMANTICS FOR SCOPED COMPUTATIONAL PATHS

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Abstract. We study how quotient topologies distinguish geometric computational paths modulo the rewrites declared by a presentation. We compare a trace-sensitive topology with one that observes endpoints, length, exact realization, and a coherent geometric representative. Duplicate labels separate these topologies. Conservative abbreviations can erase observable distinctions. A complete discrete-label presentation over the harmonic archipelago shows that geometric completeness alone does not make them agree. Equal-slot and binary timing can also give nonhomeomorphic observable based quotients. Continuous primitive word substitutions that are inverse up to scoped rewriting preserve the flat full-word global quotients and their final composable-pair domains.

The scoped quotient has continuous multiplication on the final quotient of composable representatives. We characterize agreement of this domain with the ordinary pullback of quotient arrows; the conditions imply ordinary multiplication is continuous. An open arrow projection is sufficient. For the universal presentation, a published theorem proves that the path-class projection is open under local path-connectedness and semilocal simple connectivity; this gives agreement of the pair topologies and ordinary composition continuity. Normal forms prove completeness for finite circle and torus presentations, and primitive interchange supports a product construction. For the two-circle rose, the constructed Cayley covering and tree contraction classify all continuous based loops by reduced words.

Lean checks selected results in the binary model and separately checks the equal-slot model. Structural normalization and continuous reconstruction on composable word strata identify its literal-word and tree quotients, including global arrows and final composable-pair domains. A verification map records the models, assumptions, and proof coverage.

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1. Introduction

A computational path records how an equality is established. In the judgement $a =_p b : A$, the label p names a finite derivation transforming a into b by rewrites

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and substitutions permitted by the calculus. Retaining the label makes the witness available for composition and further rewriting. The labelled account of propositional equality and its identity-type interpretation develop this perspective [2, 14].

For example, suppose that $a =_r b : A$ and $b =_s c : A$. Their composite $a =_{r;s} c : A$ records the route through b. Reversal gives $c =_{s^{-1};r^{-1}} a : A$, and the empty trace 1_a witnesses reflexivity. The traces $r; r^{-1}$ and 1_a have the same endpoints; identifying their witnesses requires a structural cancellation rule. A different witness $a =_u c : A$ is not determined by $r; s$ merely because both establish the same endpoint equality. A declared rewrite between these parallel witnesses is additional information: the endpoint judgement establishes that a equals c, whereas a rewrite between $r; s$ and u establishes equality of their witnesses.

Earlier work studies the resulting weak groupoid, whose structural laws hold through rewrites of computational paths [4]. The topological application of labelled natural deduction uses these operations to calculate fundamental groups of the circle, torus and real projective plane [3]. The book *Computational Paths: The Calculus of Equality*, by Ramos, de Queiroz, de Oliveira and Gabbay, develops the explicit-witness programme [16]. Here we study a geometric fragment whose presentation controls rewriting. Our hypotheses concern continuity and geometric soundness of the declared rules; they impose no general normalization or decision procedure on an arbitrary presentation.

In this fragment, primitive steps have continuous interval-path realizations, and each named rewrite carries an endpoint-fixed homotopy. The scoped relation is generated by these rules and the structural groupoid laws. Soundness sends scoped equality to geometric homotopy; the converse is a separate geometric-completeness problem. Parallel traces may thus remain distinct even when their realizations are homotopic. A coherent representative adds a geometric path homotopic to the realized trace. The path is data, while existence of the homotopy is a proof-irrelevant condition: the carrier stores no chosen homotopy. The construction interprets this scoped fragment without identifying type-theoretic identity proofs with continuous topological paths.

Traces are composable signed words, realized with equal time for each letter; an empty word retains its basepoint. The observable topology retains endpoints, length, exact realization and coherent path; the trace-sensitive topology also retains the signed word. Duplicate labels, conservative generator abbreviations, and a complete discrete-label presentation separate these topologies.

The presentation-invariance theorem interprets primitives by finite words of variable length. Continuity of primitive word codes, preservation of named rules, and primitive roundtrip rewrites yield homeomorphisms of trace-sensitive fixed-endpoint and global quotients and their final composable-pair domains. Binary half-and-half timing has a separate boundary: a three-label circle has nonhomeomorphic binary and equal-slot observable quotients. The Lean development preserves both models and connects the flat implementation to an independent literal-word relation and tuple-stratum topology. Homotopy of realizations supplies soundness, not topological identification.

Composition is continuous on the final quotient of composable representatives. Four equivalent quotient conditions characterize its agreement with the ordinary pair topology. An open arrow projection suffices. The published path-class open-map theorem of Holkar, Hossain, and Kulkarni [10] gives the universal positive case

under local path-connectedness and semilocal simple connectivity; the Hawaiian earring gives a discontinuous based-multiplication example. Circle and torus normal forms prove completeness, and primitive interchange constructs product presentations. An interval-glued Cayley covering and contraction classify all continuous based loops of the two-circle rose.

Section 5 discusses free topological groupoids, Brazas's τ-topology and biquotient maps; no priority claim is made for our substitution criterion or abbreviation example. The supplement retains detailed calculations, verification maps and external-input records; the full manuscript preserves the earlier exposition and distinct artifact scopes [15].

2. Geometric traces and scoped presentations

Throughout, X is a topological space and $I = [0, 1]$. Give the interval-path space X^I the compact-open topology. Let E be a topological space of oriented primitive steps, with continuous endpoint maps $s, t : E \to X$ and a continuous realization map

$$\rho : E \longrightarrow X^I, \qquad \rho(e)(0) = s(e), \quad \rho(e)(1) = t(e).$$

The continuity requirement is part of the presentation data. Let E^+ be a topological copy of E, let E^- be a formal inverse copy, and write $\widetilde{E} = E^+ \sqcup E^-$. On E^+ we use the given endpoint and realization maps; on E^- the endpoint maps are interchanged and the realization is reversed. Thus there are continuous extensions

$$\widetilde{s}, \widetilde{t} : \widetilde{E} \longrightarrow X, \qquad \widetilde{\rho} : \widetilde{E} \longrightarrow X^I,$$

whose restrictions to E^+ are s, t, ρ and whose restrictions to the inverse copy (under $E^- \cong E$) are $t, s, e \mapsto \rho(e)^{-1}$, respectively.

Figure 1 shows composition of labelled witnesses and the soundness implication from scoped rewriting to geometric homotopy. The inverse copy does not require an orientation on the topological space E: only the source, target, and direction of the represented path are reversed.

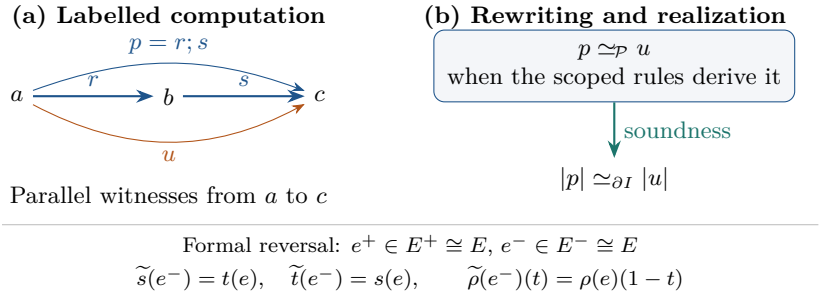


Figure 1. Labels record a route and its composition. A scoped rewrite between parallel witnesses gives an endpoint-fixed homotopy of their realizations by soundness; the converse requires geometric completeness. Formal reversal traverses a represented path in the opposite direction and leaves the topology of the step space unchanged. Endpoint maps take values in X; the realization map takes values in X^I.

Alt text: Labelled steps from a through b to c compose to p=r;s. A parallel witness u is connected to p only when the scoped rules derive their equality, which implies endpoint-fixed homotopy of their realizations. Formal reversal exchanges endpoints and reverses time.

For composable interval paths α and β and $m, n \in \mathbb{N}$, write $\alpha \star_{m,n} \beta$ for the length-weighted concatenation. When $m, n > 0$, it is given by

$$(\alpha \star_{m,n} \beta)(t) = \begin{cases} \alpha\left(\dfrac{m+n}{m}t\right), & 0 \le t \le \dfrac{m}{m+n}, \\ \beta\left(\dfrac{(m+n)t - m}{n}\right), & \dfrac{m}{m+n} \le t \le 1. \end{cases}$$

Use the natural zero-length conventions $\alpha \star_{0,n} \beta = \beta$ and $\alpha \star_{m,0} \beta = \alpha$; when $m = n = 0$, use the constant path at the common endpoint. Thus the two input paths occupy m and n equal word slots, respectively. For composable paths α, β, and γ occupying m, n, and k word slots, respectively, this gives the following literal identities, including the zero-length cases under the stated conventions:

$$(\alpha \star_{m,n} \beta) \star_{m+n,k} \gamma = \alpha \star_{m,n+k} (\beta \star_{n,k} \gamma), \qquad (\alpha \star_{m,n} \beta)^{-1} = \beta^{-1} \star_{n,m} \alpha^{-1}.$$

Example 2.1 (Reading the weights)**.** Suppose that α occupies two word slots and β occupies one. Then $\alpha \star_{2,1} \beta$ follows α until time $2/3$ and follows β on the remaining third of the interval. If γ also occupies one slot, strict reassociation is visible in the concrete identity

$$(\alpha \star_{2,1} \beta) \star_{3,1} \gamma = \alpha \star_{2,2} (\beta \star_{1,1} \gamma).$$

The weights are word-slot lengths; parentheses and homotopy witnesses are not inputs.

Definition 2.2 (geometric trace)**.** For $n \ge 1$, let $W_n \subseteq \widetilde{E}^n$ be the subspace of composable words $(e_1, \dots, e_n)$ satisfying $\widetilde{t}(e_i) = \widetilde{s}(e_{i+1})$ for every i. Put $W_0 = X$, representing empty traces, with both endpoint maps equal to the identity on X, and define the topological trace carrier by the disjoint union

$$\mathsf{Tr}_{\mathcal{P}} = \coprod_{n \in \mathbb{N}} W_n.$$

Give $\mathbb{N}$ the discrete topology. The length map is the coproduct index and is continuous. For $p = (e_1, \dots, e_n) \in W_n$ with $n \ge 1$, define its realization by equal word slots,

$$|p|(t) = \widetilde{\rho}(e_i)(nt - i + 1) \quad \text{for } 1 \le i \le n, \quad \frac{i-1}{n} \le t \le \frac{i}{n}.$$

For $p \in W_0 = X$, let $|p|$ be the constant path at its point. We write $\mathsf{Trace}(a, b)$ for the set of words with endpoints a, b. Empty words, word concatenation, and reversal are the identity, composition, and inverse operations on traces. In particular, for $p \in W_m$ and $q \in W_n$,

$$|p; q| = |p| \star_{m,n} |q|.$$

Representing traces as flat words makes the unit, associativity, double-inverse, and reversal-of-composite laws literal equalities. Their equal-slot realizations satisfy the same equalities. Inverse cancellation is different: a word followed by its reversal need not be the empty word, and its realization contracts by a homotopy.

Lemma 2.3 (continuity of trace realization)**.** *The equal-slot realization map*

$$|\cdot| : \mathsf{Tr}_{\mathcal{P}} \longrightarrow X^I$$

is continuous for the coproduct trace topology.

Proof. On each positive word stratum, evaluation is a finite pasting of continuous primitive-path evaluations. On $W_0 = X$, use the continuous constant-path map. Compactness of I transfers adjoint continuity to the compact-open path space. Detailed calculations are in the Supplement, Section S1. □

The compact-open topology also gives continuous constant-path and reversal maps on X^I. On each fixed pair of length components, the weighted operation $\star_{m,n}$ is continuous on the corresponding endpoint pullback; the length coordinates are discrete, so these operations assemble continuously over the coproduct. Word concatenation and reversal are continuous on the corresponding endpoint pullbacks of the W_n. Together with Lemma 2.3, these are the continuity facts used below. Nothing is assumed about products of quotient maps.

Let

$$\mathsf{Coh}_{\mathcal{P}}(p,\gamma) \ :\Longleftrightarrow\ \text{there exists an endpoint-fixed homotopy from } \gamma \text{ to } |p|.$$

Only the existence of such a homotopy is required; the homotopy itself is not an entry of the pair. The coherent carrier is the set

$$T = \{(p,\gamma) \in \mathsf{Tr}_{\mathcal{P}} \times X^I : \gamma(0) = s(p),\ \gamma(1) = t(p),\ \mathsf{Coh}_{\mathcal{P}}(p,\gamma)\}.$$

The trace p carries the scoped computation; γ is a chosen geometric representative observed by the topology. Coherence does not turn an ambient homotopy into a declared rewrite. The scoped quotient forgets γ within each scoped trace class, although it contributes to the raw topology.

Geometric meaning of coherence. Having the same endpoints is necessary but not sufficient for coherence. Let $A = (-1,0)$, $B = (0,1)$, and $C = (1,0)$ in $\mathbb{R}^2$. Let α traverse the two straight segments AB and BC at equal speed on the two halves of I, and put $\beta(t) = (2t-1, 0)$. In the plane,

$$H(u,t) = (1-u)\alpha(t) + u\beta(t)$$

is a continuous homotopy from α to β fixing A, C. Thus, if $|p| = \alpha$, the pair (p,β) is coherent. If instead $X = \mathbb{R}^2 \setminus \{(0, \frac{1}{3})\}$, both paths still exist, but the loop $\alpha \star_{1,1} \beta^{-1}$ surrounds the missing point once. Its winding is nonzero, so no endpoint-fixed homotopy exists in X [8]. Figure 2 depicts this change. Both paths remain continuous; what fails is that the loop between them cannot be filled inside the specified space.

More generally, for parallel paths α, β,

$$\alpha \simeq_{\partial I} \beta \quad \Longleftrightarrow \quad \alpha \star_{1,1} \beta^{-1} \simeq_{\partial I} c_{\alpha(0)}.$$

This follows from the groupoid laws of endpoint-fixed homotopy classes. Coherence is a path-space condition. Every named rewrite requires a semantic homotopy, while a homotopy alone supplies no scoped rewrite.

On this coherent carrier, define two canonical topologies. First, let

$$\kappa_{\mathrm{tr}} : T \longrightarrow \mathsf{Tr}_{\mathcal{P}} \times X^I, \qquad \kappa_{\mathrm{tr}}(p,\gamma) = (p,\gamma).$$

The *trace-sensitive topology* T_{tr} is the initial topology induced by κ_{tr}; because T is a subset of $\mathsf{Tr}_{\mathcal{P}} \times X^I$, this is just the subspace topology. In it, the complete finite word is a topological coordinate. Second, let

$$\kappa_{\mathrm{obs}} : T \longrightarrow X \times X \times \mathbb{N} \times X^I \times X^I, \qquad \kappa_{\mathrm{obs}}(p,\gamma) = (s(p), t(p), \ell(p), |p|, \gamma).$$

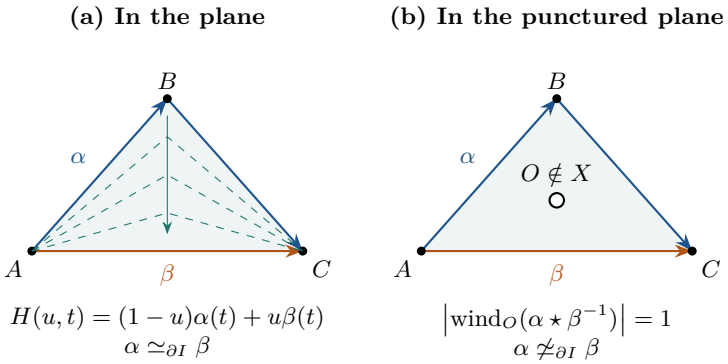


FIGURE 2. Coherence requires a homotopy inside the chosen ambient space. Left: intermediate broken paths move B to the segment AC, keeping A, C fixed. Right: deleting the interior point $O = (0, \frac{1}{3})$ prevents any such homotopy, although both boundary paths remain continuous. For $|p| = \alpha$, (p, β) is coherent only in the left-hand example. The shaded triangle is available in (a), but is not a filling in (b).

Alt text: A triangle contracts to its base segment in the plane; puncturing the triangle leaves a loop that cannot contract while avoiding the puncture.

The *observable topology* T_{obs} is the initial topology induced by κ_{obs}: it is the coarsest topology making these semantic observables continuous. Since the endpoint maps, length, and realization are continuous on the trace carrier, κ_{obs} factors continuously through κ_{tr}. Consequently the identity

$$i_T : T_{\mathrm{tr}} \longrightarrow T_{\mathrm{obs}}$$

is continuous; that is, the trace-sensitive topology is finer than the observable one.

Unless indicated otherwise, Sections 3–8 use $T = T_{\mathrm{obs}}$ and $T^{(2)} = T^{(2)}_{\mathrm{obs}}$. Sections 3, 5, 6, and 7 apply to both carrier topologies: their raw maps are continuous in both, and their descent arguments use the quotient criterion. Completeness concerns the relation alone. Section 4 checks product-quotient compatibility separately for each topology.

Write $\kappa = \kappa_{\mathrm{obs}}$. Equal observed coordinates need not imply scoped rewrite equality.

For $* = \mathrm{tr}, \mathrm{obs}$, define the common composable set with its chosen subspace topology by

$$T^{(2)}_* = \{(u, v) \in T_* \times T_* : t(u) = s(v)\}.$$

Proposition 2.4 (trace-sensitive refinement)**.** *Let T_{tr} and T_{obs} denote the two topologies above. The identity $i_T : T_{\mathrm{tr}} \to T_{\mathrm{obs}}$ is continuous, and its restriction to the common composable set is continuous:*

$$i^{(2)}_T : T^{(2)}_{\mathrm{tr}} \longrightarrow T^{(2)}_{\mathrm{obs}}.$$

Let $G^{\mathrm{tr}}_{\mathcal{P}}$ and $G^{\mathrm{obs}}_{\mathcal{P}}$ be the quotient arrow spaces for the same scoped relation, formed from the two representative topologies. Then the identity on quotient sets induces

a continuous bijection

$$J : G^{\mathrm{tr}}_{\mathcal{P}} \longrightarrow G^{\mathrm{obs}}_{\mathcal{P}}, \qquad J\, q_{\mathrm{tr}} = q_{\mathrm{obs}}\, i_T.$$

The corresponding map on final composable-pair quotients is continuous as well. In general J^{-1} need not be continuous, so J need not be a homeomorphism (Example 2.8).

Proof. The factorization of κ_{obs} through κ_{tr} proves continuity of i_T by the initial-topology criterion. Restriction to the endpoint pullback gives $i_T^{(2)}$. The identity preserves the scoped equivalence relation, so it descends to J. Since q_{tr} is a quotient map and $J\, q_{\mathrm{tr}} = q_{\mathrm{obs}}\, i_T$ is continuous, the quotient criterion proves continuity of J. The same argument applies to the componentwise final composable-pair quotients. □

Proposition 2.5 (when trace sensitivity disappears)**.** *Let $P_{\mathcal{P}} \subseteq X^I$ be the space of paths represented by coherent pairs, with its subspace topology. Write $\pi_{\mathrm{obs}} : T_{\mathrm{obs}} \to P_{\mathcal{P}}$ and $\pi_{\mathrm{tr}} : T_{\mathrm{tr}} \to P_{\mathcal{P}}$ for the maps that forget the trace. Suppose that $\mathcal{P}$ is geometrically complete and that π_{tr} has a continuous section $s : P_{\mathcal{P}} \to T_{\mathrm{tr}}$. Then the quotient comparison $J : G^{\mathrm{tr}}_{\mathcal{P}} \to G^{\mathrm{obs}}_{\mathcal{P}}$ is a homeomorphism.*

Proof. The observable code makes π_{obs} continuous. Thus $\theta = s\pi_{\mathrm{obs}} : T_{\mathrm{obs}} \to T_{\mathrm{tr}}$ is continuous. For each coherent pair u, the chosen paths of u and $\theta(u)$ are equal. Geometric completeness makes their scoped classes equal, so $q_{\mathrm{tr}}\theta = J^{-1} q_{\mathrm{obs}}$. Since q_{obs} is quotient, J^{-1} is continuous. The forward map is continuous by Proposition 2.4. □

The hypothesis is a continuous choice of trace for each represented path; completeness alone is not used to choose one. The universal one-letter section supplies such a choice. On the circle and torus based-loop presentations, a standard word selected by winding also gives a section: winding is locally constant in the compact-open based-loop space, and the geometric-path coordinate varies continuously. Thus the two quotient topologies agree on those based-loop examples.

Proposition 2.6 (completeness without trace-topology collapse)**.** *There is a geometrically complete presentation for which the comparison $J : G^{\mathrm{tr}}_{\mathcal{P}} \to G^{\mathrm{obs}}_{\mathcal{P}}$ is not a homeomorphism. On the quotient formed from its fixed-endpoint coherent carrier at one base point, the trace-sensitive topology is discrete and the observable topology is indiscrete.*

Proof. Let X be the harmonic archipelago, with base point a. Take E to be the set X^I with the discrete topology, and let each step realize the path it labels. The endpoint and realization maps are continuous because E is discrete. Declare a rewrite between every pair of parallel traces whose realizations are endpoint-fixed homotopic. Soundness and the declared rules make this presentation geometrically complete.

In the fixed-endpoint trace carrier, each trace is an isolated point: this holds for positive-length words because E is discrete, and for the empty trace because its endpoint is fixed at a. For a scoped class C, its inverse image in $T_{\mathrm{tr}}(a, a)$ is the union of the full coherent fibers over traces realizing C. Each such fiber is open in the trace-sensitive carrier. Thus every singleton is open in the based trace-sensitive quotient.

The projection $(p,\gamma) \mapsto \gamma$ from the observable based carrier to $X^I(a,a)$ is continuous. It has the continuous section $\gamma \mapsto (\langle\gamma\rangle, \gamma)$, whose observable code is $(a, a, 1, \gamma, \gamma)$. It is therefore a quotient map. Completeness identifies the scoped relation with endpoint-fixed homotopy, so the based observable quotient is homeomorphic to $\pi_1^q(X,a)$. Fabel proved that this group is uncountable and indiscrete [5, Corollaries 9–10]; the uniform loop topology used there agrees with the compact-open topology for his metric model. The based comparison is a continuous bijection from a discrete space to a non-singleton indiscrete space.

The global comparison fails as well. Choose a nonidentity class C in $\pi_1^q(X,a)$. Its inverse image under the global trace-sensitive quotient contains no empty traces and is a union of full coherent fibers over positive-length words, hence is open. If $\{C\}$ were open in the global observable quotient, its pullback along the continuous one-letter section $X^I(a,a) \to T_{\mathrm{obs}}$ would make the homotopy class C open in $X^I(a,a)$, contrary to the indiscrete quotient topology. □

Example 2.7 (Same geometry, different computation)**.** Suppose $p, q \in W_n$ have the same endpoints and the same equal-slot realization, and let γ be a common coherent representative. If $p \neq q$, then (p,γ) and (q,γ) have identical observable codes. They are therefore topologically indistinguishable in T_{obs}, even though the scoped quotient identifies them only when $p \simeq_{\mathcal{P}} q$. If the trace carrier is Hausdorff, the trace-sensitive topology T_{tr} can separate these two computationally different points. The allowed rewrites are the same in both cases; only the power of the topology to distinguish points changes.

Example 2.8 (A separation that survives the quotient)**.** Let $X = \mathbb{R}/\mathbb{Z}$ and let $E = \{e, f\}$ be discrete. Both steps are based at 0 and realize the same nonconstant loop $\lambda(t) = [t]$. Declare no named rewrites; the scoped relation contains only the structural groupoid laws. Write

$$u_e = (e, \lambda), \qquad u_f = (f, \lambda),$$

for two coherent points. Assign signed count $(1,0)$ to e and $(0,1)$ to f, extend by addition over concatenation, and negate under reversal. This map from traces to $\mathbb{Z}^2$ respects every structural groupoid law. It separates the scoped classes of u_e and u_f, despite their identical realized loops. Because the trace space retains the full word and E is discrete, the signed-count map is continuous to discrete $\mathbb{Z}^2$. It descends to a continuous map on $G_{\mathcal{P}}^{\mathrm{tr}}$, whose inverse image of $\{(1,0)\}$ is open, contains $q_{\mathrm{tr}}(u_e)$, and excludes $q_{\mathrm{tr}}(u_f)$.

Their observable codes agree. Every observable-open set containing u_e therefore contains u_f, and the same holds for their quotient classes by pullback along q_{obs}. Thus J is not a homeomorphism: Figure 3 displays the distinction that only the trace-sensitive topology retains.

Example 2.9 (One trace, two coherent representatives)**.** Let p be a trace and let γ and δ be endpoint-fixed homotopic representatives with

$$\gamma \simeq_{\partial I} |p| \simeq_{\partial I} \delta.$$

Then (p,γ) and (p,δ) are both points of T. The observable topology may observe the difference between their fifth coordinates, but the scoped quotient identifies them because the trace coordinate is the same. A coherent representative is therefore useful before quotienting without becoming extra data in a quotient arrow.

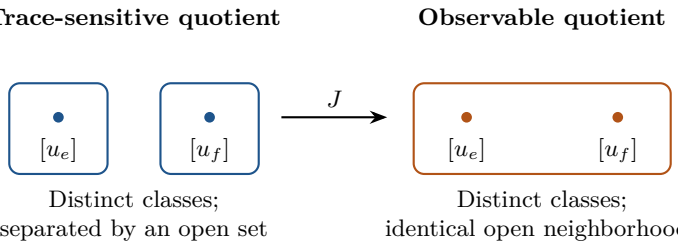


FIGURE 3. The quotient set is the same; its two topologies differ. In Example 2.8, both steps realize the same nonconstant circle loop, but their signed-count values differ. The map J preserves both scoped classes. In the observable quotient every open set containing either class contains both. Boxes illustrate separation properties only; the shared box is not asserted to be an open two-point subset.

Alt text: Two distinct scoped classes have separating neighborhoods in the trace-sensitive quotient but identical neighborhoods in the observable quotient.

For $u = (p, \gamma) \in T$, put $s(u) = s(p)$ and $t(u) = t(p)$.

Lemma 2.10 (continuity of coherent operations)**.** *For either $* = \mathrm{tr}$ or $* = \mathrm{obs}$, let $\iota : X \to T_*$ send a to the coherent identity $(1_a, c_a)$, let $\sigma : T_* \to T_*$ send (p, γ) to (p^{-1}, γ^{-1}), and let*

$$\mu_* : T_*^{(2)} \longrightarrow T_*, \qquad ((p, \gamma), (q, \delta)) \longmapsto (p; q, \gamma \star_{\ell(p),\ell(q)} \delta),$$

where the weighted operation uses the discrete trace lengths. Then all three maps are continuous.

Proof. Endpoint, length, and path coordinates are continuous under the stated operations. Full-word coordinates use concatenation and reversal on the coproduct of composable tuple strata. The zero-length cases use the constant-path formulas. Detailed calculations are in the Supplement, Section S1. □

Definition 2.11 (scoped presentation)**.** A continuous geometric rewrite presentation $\mathcal{P}$ on (X, E) consists of a predicate

$$\mathsf{Rule}_{\mathcal{P}}(p, q), \qquad p, q : \mathsf{Trace}(a, b),$$

and, for every instance of $\mathsf{Rule}_{\mathcal{P}}(p, q)$, an endpoint-fixed homotopy between the realizations of p and q.

Here $\mathsf{Rule}_{\mathcal{P}}$ specifies the named rewrites; it should not be confused with the realization comparison $R_{\mathcal{P}}$ of Section 6. The definition asks for a homotopy witness for each named rewrite and imposes no completeness condition.

Definition 2.12 (scoped rewrite equality)**.** For parallel traces, write $p \simeq_{\mathcal{P}} q$ for the inductively generated relation whose constructors are reflexivity, the named generators $\mathsf{Rule}_{\mathcal{P}}$, symmetry, transitivity, congruence under concatenation and reversal, and the following structural groupoid coherences:

$$\begin{array}{lll} (1_a); p \simeq_{\mathcal{P}} p, & p; (1_b) \simeq_{\mathcal{P}} p, & (p; q); r \simeq_{\mathcal{P}} p; (q; r), \\ p^{-1}; p \simeq_{\mathcal{P}} 1_b, & p; p^{-1} \simeq_{\mathcal{P}} 1_a, & (p^{-1})^{-1} \simeq_{\mathcal{P}} p, \\ (1_a)^{-1} \simeq_{\mathcal{P}} 1_a, & (p; q)^{-1} \simeq_{\mathcal{P}} q^{-1}; p^{-1}. & \end{array}$$

Only the rules declared by $\mathcal{P}$ and these structural constructors are available. Literal equality of traces is handled by reflexivity and substitution; ambient homotopy of their realizations is not an additional constructor.

Lemma 2.13 (weighted concatenation invariance)**.** *Let $\alpha \simeq_{\partial I} \alpha'$ and $\beta \simeq_{\partial I} \beta'$ be endpoint-fixed homotopies of composable paths. Write c_x for the constant path at x, and assume explicitly that*

$$m = 0 \Rightarrow \alpha(0) = \alpha(1) \text{ and } \alpha \simeq_{\partial I} c_{\alpha(0)},$$
$$n = 0 \Rightarrow \beta(0) = \beta(1) \text{ and } \beta \simeq_{\partial I} c_{\beta(0)},$$
$$m' = 0 \Rightarrow \alpha'(0) = \alpha'(1) \text{ and } \alpha' \simeq_{\partial I} c_{\alpha'(0)},$$
$$n' = 0 \Rightarrow \beta'(0) = \beta'(1) \text{ and } \beta' \simeq_{\partial I} c_{\beta'(0)}.$$

For arbitrary $m, n, m', n' \in \mathbb{N}$, with the zero-length conventions above,

$$\alpha \star_{m,n} \beta \simeq_{\partial I} \alpha' \star_{m',n'} \beta'.$$

Proof. For positive weights, the two piecewise-linear changes of parameter are joined by their convex interpolation, fixing endpoints. A zero-weight factor is first contracted using the corresponding null-homotopy hypothesis; when both weights vanish, contract both factors. Thus every degenerate case uses the stated geometric hypothesis. Detailed calculations are in the Supplement, Section S1. □

Theorem 2.14 (realization soundness)**.** *If $p \simeq_{\mathcal{P}} q$, then the realized paths are endpoint-fixed homotopic.*

Proof. Induct on the derivation of $p \simeq_{\mathcal{P}} q$. Reflexivity is the constant homotopy, and a named generator is sound by the corresponding presentation witness. Symmetry and transitivity use symmetry and vertical composition of endpoint-fixed homotopies. In the concatenation-congruence case, apply the weighted concatenation invariance lemma to the two inductive homotopies and the four trace lengths; this accounts for rewrites that change word length. The zero-weight hypotheses of that lemma hold because a trace of length zero is an empty word, whose realization is a constant path. For reversal, reverse the path parameter, replacing a homotopy $H(s,t)$ by $H(s,1-t)$. Unit, associativity, double-reversal, and reversal of a composite are literal equalities for the flat word realization, hence use reflexive homotopies. The two inverse-cancellation cases use the standard endpoint-fixed contraction homotopies made explicit below. This covers every rule that can occur in a scoped derivation. □

Lemma 2.15 (backtracking contracts in every space)**.** *For any continuous path $\alpha : I \to X$, the formula*

$$H(u,t) = \alpha(\min\{2t,\, 2(1-t),\, 1-u\}) \qquad (u, t \in I)$$

is an endpoint-fixed homotopy from $\alpha \star_{1,1} \alpha^{-1}$ to $c_{\alpha(0)}$.

Proof. The displayed minimum is continuous; its boundary values are the backtracking path, the constant path, and the fixed endpoints. Detailed calculations are in the Supplement, Section S1. □

Because the homotopy stays entirely in $\alpha(I)$, it requires no disk in X. For a nonempty trace p, the two factors in $p; p^{-1}$ have equal length, so its weighted realization is exactly the half-interval concatenation in Lemma 2.15; the empty case is constant. Figures 4 and 2 distinguish these two homotopy mechanisms.

Remark 2.16. The soundness theorem is one-way. Equality of realized geometric homotopy classes does not imply scoped rewrite equality unless a separate geometric completeness theorem is proved. The presentation can therefore retain computational information.

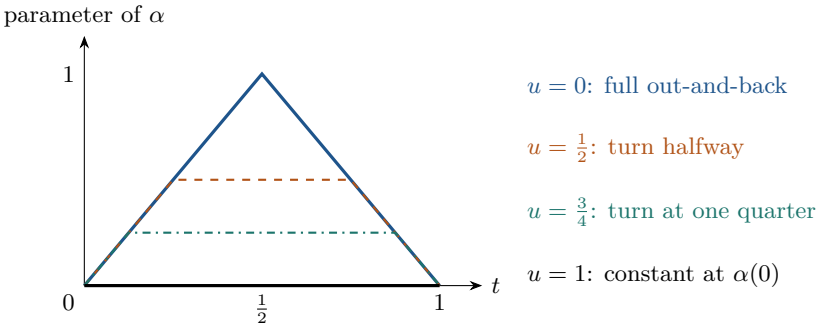


FIGURE 4. Backtracking cancellation as an explicit homotopy. The plotted function is $r_u(t) = \min\{2t, 2(1-t), 1-u\}$; the actual path is $H(u,t) = \alpha(r_u(t))$. As u increases, the turning point retreats along α and the path waits there before returning. All intermediate paths stay in $\alpha(I)$, so even a hole surrounded by α cannot obstruct this cancellation.

Alt text: The triangular parameter for an out-and-back path is capped at decreasing heights, ending at the constant path while endpoints stay fixed.

3. The quotient groupoid and its universal property

For coherent traces $u = (p, \gamma)$ and $v = (q, \delta)$, write $u \approx v$ when their endpoints agree and $p \simeq_{\mathcal{P}} q$. By realization soundness, equivalent coherent traces determine the same endpoint-fixed homotopy class of geometric paths. The quotient arrow space is

$$G_{\mathcal{P}} = T/\!\approx,$$

with the quotient topology induced by $q : T \to G_{\mathcal{P}}$. Endpoint maps descend to continuous maps

$$s, t : G_{\mathcal{P}} \longrightarrow X.$$

Reflexive paths and reversal descend because the scoped relation is closed under the corresponding constructors.

The explicit composable carrier is the $T^{(2)}$ defined in Section 2. Its quotient relation is componentwise scoped equivalence, and its quotient is denoted $G_{\mathcal{P}}^{(2),\mathrm{fin}}$. Write

$$q_{\mathrm{fin}}^{(2)} : T^{(2)} \longrightarrow G_{\mathcal{P}}^{(2),\mathrm{fin}}$$

for the quotient map. Concatenation of representatives gives a well-defined map

$$m_{\mathrm{fin}} : G_{\mathcal{P}}^{(2),\mathrm{fin}} \longrightarrow G_{\mathcal{P}}.$$

The composite $q \circ \mu : T^{(2)} \to G_{\mathcal{P}}$ is continuous and constant on equivalent pairs. The quotient criterion therefore makes m_{fin} continuous. This argument uses the quotient of composable representatives, not a product of quotient maps.

Definition 3.1 (final-domain topological groupoid)**.** Let $G \rightrightarrows X$ be a groupoid with topologies on G and X such that source, target, identity, and inverse are continuous. Give the set of composable arrow pairs $G \times_X G$ a topology, denoted C_{fin}, for which both projections $C_{\mathrm{fin}} \to G$ are continuous. We call these data a *final-domain topological groupoid* when multiplication

$$m : C_{\mathrm{fin}} \longrightarrow G$$

is continuous. The groupoid laws are the usual algebraic ones. The topology C_{fin} is part of the structure. It is at least as fine as the subspace topology from $G \times G$, and it may be strictly finer.

A discrete composable domain would satisfy this definition trivially. Our canonical domain instead has the quotient topology from explicitly composable representatives; Section 4 compares it with the ordinary subspace topology.

Theorem 3.2 (canonical final-domain groupoid)**.** *For every continuous geometric rewrite presentation $\mathcal{P}$:*

(a) s, t, the identity map $X \to G_{\mathcal{P}}$, and reversal $G_{\mathcal{P}} \to G_{\mathcal{P}}$ are continuous;
(b) m_{fin} and the two projections $G_{\mathcal{P}}^{(2),\mathrm{fin}} \to G_{\mathcal{P}}$ are continuous;
(c) the descended operations satisfy the left and right unit laws, inverse laws, and associativity.

Thus $G_{\mathcal{P}}$ is a final-domain topological groupoid.

Proof. Lemma 2.10 makes the raw operations continuous. They respect scoped equivalence, so the quotient criterion gives continuity of endpoints, identity, reversal, and m_{fin}. The projections are continuous because $\mathrm{pr}_i \circ q_{\mathrm{fin}}^{(2)} = q \circ \mathrm{pr}_i$.

Choose representatives for the algebraic laws. Flat-word units, associativity, double reversal, and reversal of a composite are literal trace and realization equalities. The inverse laws instead use the scoped constructors $p^{-1}; p \simeq_{\mathcal{P}} 1_b$ and $p; p^{-1} \simeq_{\mathcal{P}} 1_a$, sound by the endpoint-fixed backtracking contractions. Quotient soundness turns these witnesses into arrow equalities. Descent through the componentwise quotient makes the operations independent of representatives. □

A semantic interpretation must respect the rewrites and groupoid operations and forget the choice of coherent representative, as the scoped quotient does.

Theorem 3.3 (semantic factorization)**.** *Let $\mathcal{H} \rightrightarrows X$ be a topological groupoid, and let $\varphi : T \to \mathcal{H}_1$ be continuous. Assume that φ preserves endpoints, identities, reversal, and concatenation, and that every named rewrite of $\mathcal{P}$ is sent to an equality in $\mathcal{H}_1$ for every choice of coherent representatives. Assume moreover that φ is invariant under changing the coherent representative: whenever $(p, \gamma), (p, \delta) \in T$, one has*

$$\varphi(p, \gamma) = \varphi(p, \delta).$$

Then there is a unique continuous map

$$\overline{\varphi} : G_{\mathcal{P}} \longrightarrow \mathcal{H}_1$$

such that $\overline{\varphi}\, q = \varphi$. The map $\overline{\varphi}$ preserves all groupoid operations, with composition on the source interpreted on $G_{\mathcal{P}}^{(2),\mathrm{fin}}$.

Proof. Every trace p has the coherent representative $(p, |p|)$, and representative invariance gives $\varphi(p, \gamma) = \varphi(p, |p|)$ for every coherent (p, γ). It therefore suffices

to show that $p \simeq_{\mathcal{P}} q$ implies $\varphi(p, |p|) = \varphi(q, |q|)$, by induction on the derivation. Named rewrites are covered by hypothesis, and reflexivity, symmetry, and transitivity are immediate. Since $\mu((p, |p|), (q, |q|)) = (p; q, |p; q|)$ and $\sigma(p, |p|) = (p^{-1}, |p^{-1}|)$, preservation of concatenation and reversal handles the two congruence cases. The structural coherences hold because $\mathcal{H}$ satisfies the groupoid laws and φ preserves identities, reversal, and concatenation. Hence φ is constant on $\approx$. The quotient universal property gives a unique map $\overline{\varphi}$, and the quotient criterion gives continuity. The operation-preservation equations hold on representatives and hence descend. For composition, the same argument is applied to the quotient map $T^{(2)} \to G_{\mathcal{P}}^{(2),\mathrm{fin}}$; no product-quotient assumption is needed. □

3.1. **Literal words and the realization boundary.** The binary Lean carrier consists of endpoint-indexed parenthesized trees. Its binary realization recursively applies half-and-half `Path.trans`. The separate flat realization assigns each primitive letter exactly one $1/n$ slot and omits zero-length factors. Trees with the same signed word have equal flat parametrized realizations; flat and binary realizations are endpoint-fixed homotopic. The latter assertion supplies semantic soundness and does not compare observable quotient topologies.

Proposition 3.4 (literal-word correspondence)**.** *For the flat realization, flattening and canonical reconstruction identify tree scoped rewriting with the independently generated relation on composable signed words using adjacent inverse cancellation and the flattened named rules. They induce homeomorphisms of the corresponding fixed-endpoint and global scoped quotients and final composable-pair domains. For the full-word topology, these are the genuine tuple-stratum and coproduct topologies, with $W_0 = X$.*

Proof. Units, associativity, and reversal normalize each tree to the canonical reconstruction of its signed word. Adjacent cancellation has a structural witness in the tree relation. Each flattened named rule uses its named constructor, composed with structural normalization. Induction on scoped tree rewriting gives the converse. Equality of geometric paths is never used as an additional rewrite. Reconstruction is continuous on each finite tuple stratum. The induced code retains the complete signed word and the source; the source is redundant on positive strata and records the basepoint on W_0. Flattening and reconstruction descend to continuous inverse maps. Restriction to the actual endpoint-matched representative subspace gives the final-pair comparison. No ordinary-pair or global based-subspace comparison follows from this construction. □

Proposition 3.5 (fixed-endpoint coherent-representative elimination)**.** *For a fixed realization model and endpoints, give traces a topology making realization continuous and coherent representatives the topology induced by (p, γ). If scoped equivalence depends only on the trace, forgetting γ induces a homeomorphism between the coherent quotient and the trace-only scoped quotient.*

Proof. The projection $r(p, \gamma) = p$ is continuous, and $s(p) = (p, |p|)$ is a continuous section. Both maps preserve the quotient relations. We have $rs = \mathrm{id}$, while $sr(p, \gamma)$ is scoped equivalent to (p, γ) because the trace is unchanged. Quotient descent therefore gives continuous mutually inverse maps. No geometric completeness hypothesis is needed. Apply the statement separately to each realization model and each chosen trace topology. □

4. Final and ordinary composable-pair topologies

As before, $G_{\mathcal{P}} = G_{\mathcal{P}}^{\mathrm{obs}}$ and $T = T_{\mathrm{obs}}$. The definitions and theorems of this section apply equally to the trace-sensitive quotient. Compatibility for one representative topology does not imply compatibility for the other, however; the continuous map J of Proposition 2.4 does not transfer it in either direction.

There is another natural composable domain. Let

$$G_{\mathcal{P}}^{(2),\mathrm{ord}} = \{(x, y) \in G_{\mathcal{P}} \times G_{\mathcal{P}} : t(x) = s(y)\}$$

with the subspace topology. The underlying set of this space is the same as the underlying set of the final domain, but its topology need not be the same. The canonical ordinary-pair map is

$$q_{\mathrm{ord}}^{(2)} : T^{(2)} \longrightarrow G_{\mathcal{P}}^{(2),\mathrm{ord}}.$$

It is continuous and surjective. There is a unique bijection

$$j : G_{\mathcal{P}}^{(2),\mathrm{fin}} \longrightarrow G_{\mathcal{P}}^{(2),\mathrm{ord}}$$

such that

$$q_{\mathrm{ord}}^{(2)} = j \circ q_{\mathrm{fin}}^{(2)}.$$

The map j is continuous because $q_{\mathrm{fin}}^{(2)}$ is quotient. Products of quotient maps need not be quotient maps, so $q_{\mathrm{ord}}^{(2)}$ being quotient is an extra property; it does not follow formally from q being quotient.

Definition 4.1 (product-quotient compatibility)**.** The presentation has product-quotient compatibility if $q_{\mathrm{ord}}^{(2)}$ is a quotient map onto the ordinary composable-pair space. Since $q_{\mathrm{ord}}^{(2)} = j \circ q_{\mathrm{fin}}^{(2)}$ and $q_{\mathrm{fin}}^{(2)}$ is quotient, this is equivalent to j being a quotient map.

The domains have the same composable classes. Final openness is tested by pullback to $T^{(2)}$; ordinary openness comes from $G_{\mathcal{P}} \times G_{\mathcal{P}}$. Compatibility identifies these topologies (Figure 5).

Theorem 4.2 (ordinary/final comparison)**.** *For every presentation $\mathcal{P}$, the following are equivalent:*

1. *$q_{\mathrm{ord}}^{(2)} : T^{(2)} \to G_{\mathcal{P}}^{(2),\mathrm{ord}}$ is a quotient map;*
2. *$j : G_{\mathcal{P}}^{(2),\mathrm{fin}} \to G_{\mathcal{P}}^{(2),\mathrm{ord}}$ is a quotient map;*
3. *the inverse bijection $j^{-1} : G_{\mathcal{P}}^{(2),\mathrm{ord}} \to G_{\mathcal{P}}^{(2),\mathrm{fin}}$ is continuous;*
4. *j is a homeomorphism, equivalently the final and ordinary composable-pair topologies coincide.*

Thus each condition is also equivalent to product-quotient compatibility. When these conditions hold, multiplication is continuous from the ordinary pullback domain. Theorem 3.2 uses none of these conditions.

Proof. If j is quotient, so is the composite $q_{\mathrm{ord}}^{(2)} = j \circ q_{\mathrm{fin}}^{(2)}$. Conversely, suppose $q_{\mathrm{ord}}^{(2)}$ is quotient and $j^{-1}(U)$ is open. Then $(q_{\mathrm{ord}}^{(2)})^{-1}(U) = (q_{\mathrm{fin}}^{(2)})^{-1}(j^{-1}(U))$ is open, so U is open. This proves that (1) and (2) are equivalent.

The map j is a continuous bijection. A bijective quotient map has continuous inverse, so (2) and (3) are equivalent; together with the already known continuity of j, these are equivalent to (4). The homeomorphism condition is precisely equality

of the two topologies on the common underlying set. Finally, when these conditions hold, multiplication is continuous by the quotient criterion applied to its continuous representative-level concatenation. □

Remark 4.3 (terminology). In the standard definition of a topological groupoid, multiplication is continuous on the ordinary pullback $G_{\mathcal{P}} \times_X G_{\mathcal{P}}$. Accordingly, the phrase "topological groupoid" without qualification is reserved for the case in which product-quotient compatibility has been proved. The unconditional object of Theorem 3.2 is instead called a *final-domain topological groupoid*: it has the same arrow set and groupoid laws, but its multiplication domain carries the final topology from explicitly composable representatives.

Remark 4.4 (comparison with convenient categories). One can instead change the ambient category of spaces, for example by using compactly generated spaces and their k-ified products [11]. That remedy changes the topology assigned to the ordinary product. The present construction stays in Top and records the final topology forced by the chosen representative quotient, so the failure of the ordinary product remains visible. We do not claim that the two remedies agree in general.

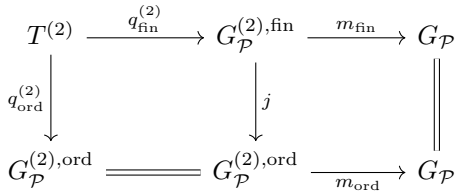


FIGURE 5. The composition-continuity boundary. The upper multiplication is always continuous. The comparison j is always a continuous bijection, and is a homeomorphism exactly under Theorem 4.2's equivalent quotient conditions. These conditions imply continuity of m_{ord}. The paper does not prove the converse, that continuity of m_{ord} alone implies them.

Alt text: The same composable arrows carry a final and an ordinary topology. Their continuous bijection is a homeomorphism exactly under product-quotient compatibility.

Proposition 4.5 (open-map criterion)**.** *If the bijection from the final composable quotient to the ordinary pair space is an open map, then product-quotient compatibility holds. In particular, an ordinary-pair map $q^{(2)}_{\mathrm{ord}}$ that is open is quotient, so multiplication is continuous for the ordinary pullback topology.*

Proof. An open continuous bijection is a homeomorphism, so Theorem 4.2 applies. An open continuous surjection is a quotient map. □

Theorem 4.6 (open arrow quotients)**.** *If the arrow quotient $q : T \to G_{\mathcal{P}}$ is open, then $q^{(2)}_{\mathrm{ord}} : T^{(2)} \to G_{\mathcal{P}}^{(2),\mathrm{ord}}$ is an open quotient map. Consequently the final and ordinary composable-pair topologies agree, and multiplication is continuous on the ordinary pullback.*

Proof. The product $q \times q : T \times T \to G_{\mathcal{P}} \times G_{\mathcal{P}}$ is an open surjection. Because scoped equivalence preserves endpoints, $T^{(2)}$ is the full inverse image of $G_{\mathcal{P}}^{(2),\mathrm{ord}}$ under this

product map. Restricting an open map to the full inverse image of a subspace gives an open map onto that subspace: the image of $U \cap T^{(2)}$, for U open in $T \times T$, is $(q \times q)(U) \cap G_{\mathcal{P}}^{(2),\mathrm{ord}}$. This restriction is $q_{\mathrm{ord}}^{(2)}$, and it is continuous and surjective. It is therefore quotient, so Theorem 4.2 applies. □

Theorem 4.7 (compact-Hausdorff compatibility)**.** *Suppose that the final composable domain $G_{\mathcal{P}}^{(2),\mathrm{fin}}$ is compact and that the ordinary composable domain $G_{\mathcal{P}}^{(2),\mathrm{ord}}$ is Hausdorff. Then product-quotient compatibility holds. Hence the final and ordinary composable-pair topologies agree and ordinary-pullback multiplication is continuous.*

Proof. The comparison map

$$j : G_{\mathcal{P}}^{(2),\mathrm{fin}} \longrightarrow G_{\mathcal{P}}^{(2),\mathrm{ord}}$$

is a continuous bijection. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism: its image of every closed set is compact, hence closed, so its inverse is continuous. The composite $j \circ q_{\mathrm{fin}}^{(2)} = q_{\mathrm{ord}}^{(2)}$ is therefore a quotient map. This is exactly product-quotient compatibility. □

Corollary 4.8 (discrete arrow spaces)**.** *If $G_{\mathcal{P}}$ is discrete, then the final and ordinary composable-pair topologies coincide, and both are discrete.*

Proof. The ordinary domain is a subspace of the discrete space $G_{\mathcal{P}} \times G_{\mathcal{P}}$, so it is discrete. Every subset of the final domain is then the preimage under the continuous map j of an open set, hence open. So both topologies are discrete and j is a homeomorphism. □

In both results, Theorem 4.2 turns agreement of the two topologies into continuity of ordinary multiplication. Theorem 3.2 needs neither hypothesis.

4.1. **A strict failure: the Hawaiian earring.** Let

$$\mathbb{H} = \bigcup_{n \geq 1} \left\{ (x, y) \in \mathbb{R}^2 : \left(x - \frac{1}{n} \right)^2 + y^2 = \frac{1}{n^2} \right\}$$

be the Hawaiian earring, based at $x_0 = (0, 0)$. Write $\Omega = \Omega(\mathbb{H}, x_0)$ for its based loop space with the compact-open topology and

$$q_\Omega : \Omega \longrightarrow \pi_1^q(\mathbb{H}, x_0)$$

for the endpoint-fixed homotopy quotient. Fabel proved that multiplication in $\pi_1^q(\mathbb{H}, x_0)$ is not continuous [6]. Loop concatenation $\Omega \times \Omega \to \Omega$ is continuous and induces this multiplication, so $q_\Omega \times q_\Omega$ is not a quotient map either.

Proposition 4.9 (Hawaiian-earring obstruction)**.** *Use the observable based fiber of the universal presentation and the fiberwise quotient construction of Lemma 7.2:*

$$T_0 := T_{\mathcal{U}_{\mathbb{H}}}^{\mathrm{obs}}(x_0, x_0), \qquad G_0 := G_{\mathcal{U}_{\mathbb{H}}}^{\mathrm{obs}}(x_0, x_0),$$

with quotient map $q_0 : T_0 \to G_0$ and projection $\pi_0 : T_0 \to \Omega$. Then the ordinary pair map

$$q_0 \times q_0 : T_0 \times T_0 \longrightarrow G_0 \times G_0$$

is not quotient. Consequently the final topology on composable pairs is strictly finer than the ordinary product topology, and multiplication on the ordinary product is discontinuous.

Proof. The fiberwise lemma gives that π_0 is quotient and has the continuous section $\sigma(\gamma) = (\langle\gamma\rangle, \gamma)$. It also gives a fiberwise universal-completion homeomorphism

$$h : G_0 \longrightarrow \pi_1^q(\mathbb{H}, x_0), \qquad h \circ q_0 = q_\Omega \circ \pi_0.$$

Suppose that $q_0 \times q_0$ were quotient. Postcomposition with the homeomorphism $h \times h$ would make

$$(q_\Omega \times q_\Omega) \circ (\pi_0 \times \pi_0)$$

a quotient map. If $(q_\Omega \times q_\Omega)^{-1}(U)$ were open, its preimage under the continuous map $\pi_0 \times \pi_0$ would therefore be open; the quotient criterion for the displayed composite would force U to be open. Hence $q_\Omega \times q_\Omega$ would be quotient, contradicting Fabel's theorem.

Theorem 4.2 and its proof apply to the based fiber without change, with $T_0 \times T_0$ in place of $T^{(2)}$. They show that the continuous bijection from the final pair domain to the ordinary one is not a homeomorphism. Since the final topology is always at least as fine, it is strictly finer here. Finally, h preserves loop multiplication, so continuity of ordinary-product multiplication on G_0 would imply continuity on $\pi_1^q(\mathbb{H}, x_0)$, again contradicting Fabel's theorem. □

Proposition 4.9 is stated for the observable quotient used in Theorem 4.2. For this universal presentation, Corollary 7.4 also applies on the based carrier: the one-letter path section is continuous for the trace-sensitive topology and recovers the same homotopy class. Its product with itself is a continuous section on based composable representatives. Repeating the product-quotient argument above with that section transfers the strict final/ordinary pair distinction and the discontinuity of ordinary multiplication to the trace-sensitive based quotient. The general refinement map alone would not establish that conclusion. The pair-section argument here is mathematical; the formal selection of the universal-collapse module checks the arrow comparison.

The discontinuity result is Fabel's. We use it to show that the two composable-pair topologies do differ, and that the failure sits at the inverse of the comparison map j. With the final topology, multiplication is continuous here as it is for every presentation, while for the ordinary product topology it is not.

5. Functoriality

Let $\mathcal{P}$ and $\mathcal{Q}$ be presentations on (X, E) and (Y, F). A presentation map consists of a continuous map $f : X \to Y$, a continuous step map preserving endpoints and realizations, and a proof that every named rule of $\mathcal{P}$ maps to a scoped derivation in $\mathcal{Q}$. Presentation maps form a category: the identity has the identity space and step maps, and composition is the composite of the corresponding space and step maps, with the scoped-derivation proofs composed by induction.

Theorem 5.1 (functoriality)**.** *Every presentation map $F : \mathcal{P} \to \mathcal{Q}$ induces a continuous map*

$$F_1 : G_\mathcal{P} \longrightarrow G_\mathcal{Q}$$

that preserves source, target, identities, reversal, and final-domain composition. The assignment preserves identity maps and composites.

Proof. Write $f : X \to Y$ for the underlying map of spaces and F_*p for the trace obtained by applying the step map to every letter of p. The raw map is

$$\widetilde{F} : T_{\mathcal{P}} \longrightarrow T_{\mathcal{Q}}, \qquad (p, \gamma) \longmapsto (F_*p, f \circ \gamma).$$

It preserves coherence because realizations are preserved by the step map and endpoint-fixed homotopies may be postcomposed with f. It is continuous: on every word stratum the induced trace map is continuous, postcomposition $X^I \to Y^I$ is continuous for the compact-open topology, and the observable code of $\widetilde{F}(p, \gamma)$ is consequently a continuous function of the observable code of (p, γ). Induction on scoped rewrite derivations proves that $\widetilde{F}$ sends related traces to related traces. It therefore descends through the quotient, and the quotient criterion gives continuity on arrows. Applying the step map to reflexivity, reversal, and concatenation gives the corresponding preservation equations before quotienting. The same argument on the explicit composable carrier gives a continuous map on final composable domains and proves final composition preservation. Identity and composite step maps are definitionally the identity and composite maps on raw traces; quotient extensionality completes the proof. □

The same construction gives ordinary-pair continuity for every presentation map. If both source and target presentations satisfy product-quotient compatibility, the ordinary multiplication maps are continuous as well. Continuity of the induced map on arrows does not require this additional compatibility assumption. For the trace-sensitive topology, the word component of $\widetilde{F}$ is continuous on each coproduct stratum and the path component is the same postcomposition map, so the proof carries over.

5.1. **Parametrization dependence.** Figure 6 compares the two execution schedules.

Proposition 5.2 (binary timing can identify observable codes)**.** *There is a three-generator based circle presentation with no named rewrite rules for which the observable quotients of the fixed-base carriers differ: the binary quotient is not T_0, and the equal-slot flat quotient is discrete.*

Proof. Let $\alpha(t) = [t]$ in $\mathbb{R}/\mathbb{Z}$, and let the primitive labels a, b, c realize α, $\beta = \alpha \star \alpha$, and $\chi = \alpha \star \beta$, respectively, with $\star$ denoting ordinary half-and-half concatenation. The binary traces $p = (a; b); b$ and $q = c; (a; a)$ both realize $\chi \star \beta$. They have length three and can choose this same path as coherent representative, so their binary observable codes agree. Signed occurrence count of c is invariant under structural scoped rewrites and has values zero and one on these two traces. Their scoped classes are distinct. Every open set in the induced raw observable topology contains both chosen representatives or neither; the same holds for their classes by the definition of quotient topology. Hence that quotient is not T_0.

For the flat construction, the six oriented primitive paths have distinct winding numbers $1, 2, 3, -1, -2, -3$. Restriction to each prescribed equal slot therefore recovers the signed letter. At each fixed length there are finitely many words, with distinct realizations in the Hausdorff compact-open path space. This finite image is discrete, so the fiber of each observed word is open. The discrete length coordinate separates the strata. A scoped-class preimage is a union of word fibers, even when the coherent path varies; thus the fixed-base flat quotient is discrete. This

last assertion concerns the quotient of the based carrier, not an unproved subspace comparison with the global quotient. □

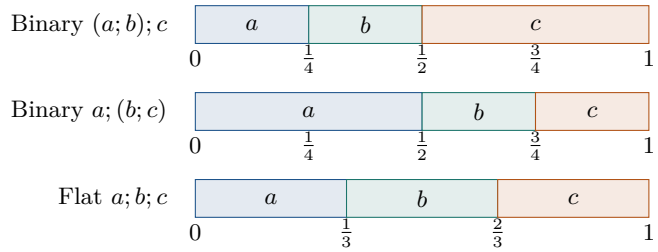


FIGURE 6. Exact timing depends on the trace model. Recursive binary concatenation halves each concatenation interval; equal-slot realization assigns one third to each letter of this word, independently of parentheses. Homotopy between these paths does not identify topologies that observe their exact parametrizations.

Alt text: Two parenthesizations of three letters have binary durations one quarter, one quarter, one half and one half, one quarter, one quarter. The flat model gives every letter one third.

5.2. **Finite-word interpretations and a conservative abbreviation.** A primitive may instead be interpreted as a finite trace with the same endpoints. Extend such a substitution M through identities, composition, and reversal. If images of every named source rule are target scoped rewrites, induction proves preservation of the entire scoped relation. Primitive roundtrip rewrites for M, N then extend to every trace. This is the algebraic part of a presentation change. It does not by itself establish continuity when length and exact realization are observable.

The fixed-endpoint criterion and the observable counterexample use endpoint-indexed *tree* trace carriers $\mathcal{T}_{\mathcal{P}}(a,b)$ and their fixed-endpoint scoped quotients $\mathcal{Q}_{\mathcal{P}}(a,b)$, with the stated trace topologies. Their Lean claims therefore do not assume the literal-word correspondence discussed in Section 9.

Proposition 5.3 (fixed-endpoint substitution criterion)**.** *Let M, N be endpoint-preserving finite-trace substitutions in opposite directions, preserving named rules. Suppose their primitive roundtrips are scoped rewrites to the corresponding primitive and their recursively extended trace maps are continuous for the supplied trace topologies. They induce mutually inverse homeomorphisms of the fixed-endpoint scoped trace quotients. For binary observable trace topologies, continuity of either substitution follows if all substituted primitives have the same length k and their binary realizations equal the corresponding source primitive paths exactly.*

Proof. Induction proves preservation of scoped rewrites and extends the primitive roundtrip witnesses to all traces. Quotient descent gives inverse maps and the quotient criterion gives their continuity. For the binary specialization, induction gives

$$\ell(Mp) = k\ell(p), \qquad |Mp|_{\mathrm{bin}} = |p|_{\mathrm{bin}}.$$

The output observation is thus the continuous function $(n,\delta) \mapsto (kn,\delta)$ of the input observation. Apply this argument in both directions, with their respective expansion lengths. □

The hypotheses concern the actual substitutions; no quotient homeomorphism is an input. The general criterion is quotient-topology infrastructure. The binary specialization derives continuity from primitive data, but its uniform-length and exact-timing conditions are restrictive. For the flat full-word topology, the following criterion instead permits variable-length substitutions, including empty words.

Theorem 5.4 (presentation invariance from primitive word codes)**.** *Let M, N be endpoint-preserving finite-trace substitutions between presentations over the same space X. Suppose they preserve named rules and their primitive roundtrips are scoped rewrites to the original primitives. Suppose also that the two maps sending a primitive to the full signed word of its substituted trace are continuous into the coproduct of finite word spaces. Then they induce homeomorphisms of the flat trace-sensitive scoped quotients at fixed endpoints, of global arrow quotients, and of the final composable-pair domains. The pair maps commute with the two arrow projections. For discrete primitive alphabets the continuity hypotheses hold automatically.*

Proof. On each finite input stratum, substitution concatenates finitely many output words. Continuity of the primitive word codes makes this operation continuous; the discrete length coordinate isolates each output-length stratum. Equal-slot execution is continuous on the compatible finite-product strata, including the empty stratum X. Thus the output full word, endpoints, and realized path are continuous functions of the input source and full word. Choosing that realized path as the coherent representative gives a continuous map of the global carriers. Restricting its product to matching endpoints gives the continuous map of actual composable representative subspaces. Induction extends the primitive roundtrip rewrites to arbitrary traces. Quotient descent therefore gives inverse continuous maps in all three cases. The pair projection equations already hold on representatives. □

The continuity conclusion is derived from primitive data; continuity of the whole substitution or a quotient homeomorphism is not assumed. These homeomorphisms preserve endpoints and scoped computation. Compatibility with the geometric homotopy comparison requires the additional condition that each substituted primitive realization is homotopic to the original path; `WordSubstitution.map_flat_homotopic` verifies its extension to all traces. Neither statement identifies final and ordinary composable topologies.

For a nondiscrete application, take $X = \mathbb{R} \times \mathbb{R}/\mathbb{Z}$ and the primitive family $E = \mathbb{R}$, with e_x based at $(x, 0)$ and realizing $t \mapsto (x, [t])$. Declare no named rules. Enlarge the alphabet to $E' = \mathbb{R} \sqcup \mathbb{R}$ by adding a_x, realizing $t \mapsto (x, [2t])$, with the defining rule $a_x \simeq_{\mathcal{P}} e_x; e_x$. The realization families are continuous, and the defining rule is sound by exact equal-slot realization. Inclusion and elimination $a_x \mapsto e_x; e_x$ preserve the rules; their primitive roundtrips are identities or instances of the defining rule. The elimination word code is the continuous map $x \mapsto (e_x)$ on one component and the diagonal map $x \mapsto (e_x, e_x)$ on the other. Inclusion is continuous into the one-letter stratum. Theorem 5.4 therefore gives the global and final-pair homeomorphisms for these nondiscrete alphabets. Output length is one on the old component and two on the added component. In general, continuous word codes force output length to be locally constant, hence constant on a connected primitive alphabet.

Figure 7 illustrates the following conservative move.

Proposition 5.5 (a redundant generator erases an observable distinction)**.** *There is an algebraically conservative abbreviation move whose canonical fixed-endpoint quotient equivalence has a discontinuous inverse for the equal-slot observable trace topologies.*

Proof. Choose any based path α. The old discrete-label presentation has primitives e, f, realizing $\alpha, \alpha \star \alpha$, and no named rules. Length parity $\ell(p)$ mod 2 is invariant under structural scoped rewrites: cancellation removes an even number of letters. It is continuous because length is an observed discrete coordinate, and therefore descends to a continuous quotient invariant with values 1 on $[f]$ and 0 on $[e; e]$.

Add a primitive a realizing $\alpha \star \alpha$ and the sole defining rule $a \simeq_{\mathcal{P}} e; e$. Inclusion and substitution $a \mapsto e; e$ preserve all named rules and have primitive roundtrips equal up to scoped rewriting. They consequently induce inverse algebraic quotient maps. In the expanded presentation, the one-letter traces a, f have identical flat observable coordinates. Their quotient classes are inseparable. They are distinct: signed count of f is preserved by all structural rules and by the defining rule, and takes values $0, 1$. Elimination sends these inseparable classes to $[e; e], [f]$, whose old continuous parity values differ. If elimination were continuous, its composite with parity would send inseparable points to distinct points of a discrete space, a contradiction. □

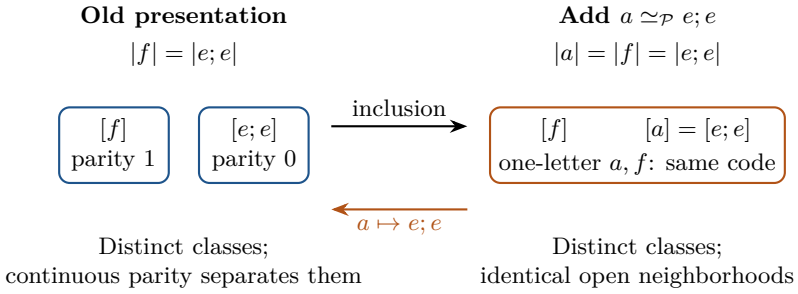


FIGURE 7. A primitive-to-word substitution preserves the presented algebra but can change the observable topology (Proposition 5.5). Adding the defining rule gives inverse scoped substitutions. Inclusion is continuous on the observable quotient; elimination is discontinuous because it sends inseparable classes to classes separated by parity. Boxes indicate separation properties, not asserted open two-point subspaces.

Alt text: Old classes f and e;e are separated by length parity. After adding a, its class equals e;e, while the one-letter representatives a and f share an observable code. Eliminating a sends inseparable classes to separated old classes.

This checked statement concerns the canonical presentation equivalence; it does not claim nonhomeomorphism under every unrelated bijection. It needs neither a nontrivial geometric loop nor geometric completeness. Matching coherent-representative elimination transports this fixed-endpoint result to the corresponding coherent quotients. Henry and Mimram study Tietze equivalence for monoid presentations [9]. Mimram and Oleon characterize when 1-polygraphs present equivalent sets in homotopy type theory and transfer coherence structures

along their Tietze transformations [13]. Here the additional question is continuity for specified quotient topologies on geometric representatives: scoped roundtrips establish the algebraic equivalence, while primitive word codes control the full-word topology and exact observations can obstruct continuity. Brazas uses free topological groups to obtain a topological-group topology on the fundamental group [1]. Our final composable domain retains the declared arrow quotient topology. No priority claim is made for the abbreviation example or substitution criterion.

Hardy and Morris characterize free topological groupoids by extension of continuous graph morphisms, with composition continuous on the ordinary composable subspace [7]. Our construction adds declared rewrites and exact geometric observations and initially gives composable classes a final topology. Identifying it with their universal construction would require a separate argument. Michael's biquotient maps include open surjections and are stable under products and restriction over codomain subspaces [12]. They suggest a broader sufficient condition for ordinary/final agreement than the open-arrow criterion. That generalization is not proved here; product closure should not be confused with stronger characterizations that require additional hypotheses on the range.

6. Geometric comparison and complete presentations

Let H_{geo} be the quotient of coherent traces by equality of endpoints and endpoint-fixed homotopy of chosen geometric representatives, and write

$$h : T \longrightarrow H_{\mathrm{geo}}$$

for the quotient map. Define the final composable geometric domain by

$$H_{\mathrm{geo}}^{(2),\mathrm{fin}} = T^{(2)}/\!\sim_{\mathrm{geo}}^{(2)}, \qquad (u,v) \sim_{\mathrm{geo}}^{(2)} (u',v') \Longleftrightarrow h(u) = h(u') \text{ and } h(v) = h(v').$$

Weighted concatenation descends continuously to this final domain. Its zero-weight hypotheses hold because a coherent representative of an empty trace is null-homotopic. Identity and reversal likewise descend, giving H_{geo} its final-domain groupoid structure.

The realization comparison and its composable-domain map are

$$R_{\mathcal{P}} : G_{\mathcal{P}} \longrightarrow H_{\mathrm{geo}}$$

and

$$R_{\mathcal{P}}^{(2)} : G_{\mathcal{P}}^{(2),\mathrm{fin}} \longrightarrow H_{\mathrm{geo}}^{(2),\mathrm{fin}}, \qquad ([u],[v]) \longmapsto ([u]_{\mathrm{geo}}, [v]_{\mathrm{geo}}).$$

The target contains represented geometric classes, not every ambient path class: its surjectivity is built into the quotient of T. The induced map $H_{\mathrm{geo}} \to \Pi_1^q(X)$ is a continuous injection, not asserted to be an embedding. The universal presentation represents all paths; the finite circle and torus presentations cover all based-loop classes at their chosen basepoints. Soundness makes $R_{\mathcal{P}}$ well-defined; completeness identifies the two relations on T, hence their quotient topologies.

Theorem 6.1 (comparison morphism)**.** *The map $R_{\mathcal{P}}$ is continuous, surjective on the geometric quotient, and preserves source, target, identity, and reversal. The induced pair map $R_{\mathcal{P}}^{(2)}$ is continuous, and the two maps commute with composition on the final domains. In particular, $R_{\mathcal{P}}$ is a continuous groupoid morphism for the final-domain structures.*

Proof. Soundness makes the comparison well-defined; $h = R_{\mathcal{P}} \circ q$ and the quotient criterion give continuity. Surjectivity follows from the definition of H_{geo}. Raw identity and reversal give their preservation equations. Weighted concatenation invariance gives composition preservation. The continuous raw pair quotient is constant on componentwise scoped equivalence, so descent through $q^{(2)}_{\mathrm{fin}}$ gives the continuous pair comparison. □

Definition 6.2 (geometric completeness)**.** The presentation is geometrically complete if, for all coherent representatives u, v, equality of their geometric homotopy classes implies scoped equivalence:

$$R_{\mathcal{P}}(q(u)) = R_{\mathcal{P}}(q(v)) \quad \Longrightarrow \quad u \approx v.$$

Theorem 6.3 (faithfulness and homeomorphism)**.** *The comparison $R_{\mathcal{P}}$ is faithful, equivalently injective on arrows because its object map is the identity, if and only if $\mathcal{P}$ is geometrically complete. If $\mathcal{P}$ is geometrically complete, then*

$$R_{\mathcal{P}} : G_{\mathcal{P}} \xrightarrow{\cong} H_{\mathrm{geo}}$$

is a homeomorphism on arrow spaces and an isomorphism of abstract groupoids.

Proof. If the comparison is injective, equality of the geometric homotopy classes of two coherent representatives gives equality of their scoped classes. This is precisely geometric completeness. Conversely, completeness turns equality of comparison images into equality of scoped classes. The comparison is always surjective and continuous. Under completeness, let $h : T \to H_{\mathrm{geo}}$ be the geometric quotient map. If $h(u) = h(v)$, completeness gives $u \approx v$, hence $q(u) = q(v)$. Therefore q factors uniquely through a map

$$S : H_{\mathrm{geo}} \longrightarrow G_{\mathcal{P}}, \qquad S \circ h = q.$$

Since h is quotient and q is continuous, S is continuous. The two composites are identity maps by quotient induction. □

In the trace-sensitive version, $G_{\mathcal{P}}$ and H_{geo} both carry the quotient topologies from T_{tr}. The homeomorphism always compares two quotients of the same carrier.

Remark 6.4 (Two distinct obligations). Geometric completeness concerns *which arrows are equal*: it compares the scoped and geometric equivalence relations. Product-quotient compatibility concerns *which composable-pair sets are open*: it compares the final and ordinary domain topologies and permits continuity of multiplication to pass to the ordinary pullback. The former does not imply the latter. The universal presentation is geometrically complete for every X, yet in the Hawaiian-earring example of Proposition 4.9 multiplication on its observable based-fiber quotient is discontinuous for the ordinary product topology. This also obstructs global compatibility. The base point is closed in the Hausdorff Hawaiian earring, so continuity of source and target makes its based arrow fiber closed. A quotient map restricted over a closed target subset remains quotient: a closed restricted preimage is closed in the whole domain, and the quotient criterion then gives closedness in the target. Thus the based quotient has the global fiber's subspace topology. If the global ordinary pair map were quotient, its restriction over the closed based-pair subset would be quotient as well, contradicting Proposition 4.9.

Definition 6.5 (normal-form certificate)**.** A normal-form certificate for $\mathcal{P}$ consists of a set NF of normal-form codes, a representative map $\nu : \mathsf{NF} \to T$, and a normalizer $N : T \to \mathsf{NF}$ such that, for all coherent representatives u, v,

(a) $u \approx \nu(N(u))$; and
(b) $h(u) = h(v)$ implies $N(u) = N(v)$.

Clause (a) requires a scoped derivation to the chosen normal representative. Clause (b) requires geometrically equivalent representatives to receive the same code; equivalently, representatives with different codes lie in different geometric classes. A code records only the normal form; it stores neither a chosen homotopy nor a proof of normalization. No further condition on the representatives is needed, because equal codes have equal images under ν.

Theorem 6.6 (normal-form completeness criterion)**.** *A normal-form certificate implies geometric completeness. Therefore the realization comparison is faithful and is a homeomorphism on arrow spaces.*

Proof. Assume $h(u) = h(v)$. Clause (b) gives $N(u) = N(v)$. Combining this with clause (a) for u and the symmetric form of clause (a) for v yields

$$u \approx \nu(N(u)) = \nu(N(v)) \approx v.$$

This is geometric completeness, and the faithfulness and homeomorphism claims follow from the comparison theorem. Figure 8 shows the two obligations separately. Neither N nor ν needs to be continuous. The certificate is a set-level object, and the topological conclusion comes from quotient factorization. □

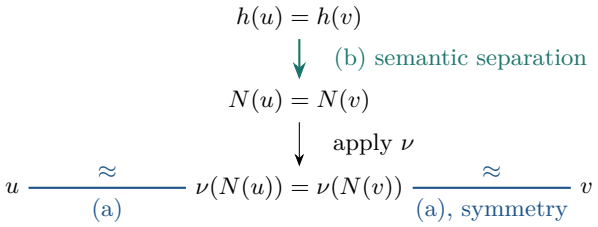


FIGURE 8. A normal-form certificate proves completeness by two different means: syntactic normalization (a) connects each trace to its chosen normal representative; semantic separation (b) forces the two codes to agree. The bottom row is then scoped transitivity, yielding $u \approx v$. The arrows are logical steps and assume nothing about continuity of N or ν.

Alt text: Syntactic normalization connects a trace to its normal form; geometric separation says homotopic normal forms have the same index.

Corollary 6.7 (fiberwise normal-form completeness)**.** *For fixed endpoints $a, b \in X$, the same definitions and proof apply to $T(a, b) \subseteq T$. In particular, a normal-form certificate on $T(a, a)$ proves geometric completeness of the based-loop comparison on that fiber.*

Proof. All endpoint maps and the scoped relation preserve fixed endpoint fibers, so the two normal-form clauses and the proof of the criterion restrict to $T(a, b)$ without change. □

7. The universal presentation

For every topological space X, take the primitive step space to be the continuous interval path space X^I. Its endpoint maps are evaluation at 0 and 1. Declare two parallel traces to be named generators exactly when their realized paths are endpoint-fixed homotopic. Thus every continuous path γ is itself available as a one-letter primitive trace. If two realizations are endpoint-fixed homotopic, that homotopy is literally one of the declared universal rules. The universal presentation is therefore complete in one step: the ambient homotopy relation is built into the presentation instead of being derived from a normalizer.

The two lemmas below treat both representative topologies explicitly. Without a superscript, $T_{\mathcal{U}_X}$ and $G_{\mathcal{U}_X}$ refer to the observable choice.

Lemma 7.1 (coherent-path quotient identification)**.** *For the universal presentation, the projection*

$$\pi : T_{\mathcal{U}_X} \longrightarrow X^I, \qquad \pi(p,\gamma) = \gamma,$$

is a continuous quotient map. Consequently, the quotient of $T_{\mathcal{U}_X}$ by endpoint-fixed homotopy of the chosen representative is canonically homeomorphic to the usual quotient-topologized path-class space

$$\Pi_1^q(X) = X^I/{\simeq_{\partial I}},$$

with the endpoint maps retained.

Proof. The one-letter section is continuous for both topologies, because the primitive path coordinate has the compact-open topology. It is inverse after quotienting: universal scoped equivalence is exactly endpoint-fixed homotopy. Detailed calculations are in the Supplement, Section S2. □

Lemma 7.2 (fiberwise coherent-path quotient)**.** *Fix $a, b \in X$, and let*

$$X^I(a,b) = \{\gamma \in X^I : \gamma(0) = a,\ \gamma(1) = b\}$$

carry its subspace topology. For $* = \mathrm{tr}, \mathrm{obs}$, define the coherent carrier with these fixed endpoints

$$T^*_{\mathcal{U}_X}(a,b) = \{(p,\gamma) \in T^*_{\mathcal{U}_X} : s(p) = a,\ t(p) = b\}$$

with its own subspace topology, and define $G^*_{\mathcal{U}_X}(a,b)$ to be its quotient by scoped equivalence, with the quotient topology. The restricted projection

$$\pi^*_{a,b} : T^*_{\mathcal{U}_X}(a,b) \longrightarrow X^I(a,b)$$

is quotient. Hence $G^*_{\mathcal{U}_X}(a,b)$ is canonically homeomorphic to the endpoint-fixed homotopy quotient of $X^I(a,b)$.

Proof. Restrict the continuous projection and one-letter section to the fixed endpoint fibers. Their composites agree with the identity after the specified geometric quotient. Detailed calculations are in the Supplement, Section S2. □

Figure 9 records both requirements: a continuous section makes representative projection quotient, and agreement of the scoped and homotopy relations identifies the quotient spaces.

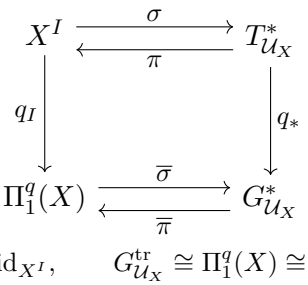


FIGURE 9. The universal one-letter section supplies the topology, and the universal rules supply the equivalence relation. Here $* \in \{\mathrm{tr}, \mathrm{obs}\}$ and $\sigma(\gamma) = (\langle\gamma\rangle, \gamma)$; q_I is the homotopy quotient map. The lower maps are induced by σ and π. The upper maps are continuous, with $\pi\sigma = \mathrm{id}$, so π is quotient. Agreement of scoped and homotopy relations makes the lower maps mutually inverse homeomorphisms. Under these identifications, $J_{\mathcal{U}_X}$ is the identity on path classes.

Alt text: A continuous one-letter section splits representative projection; matching scoped and homotopy relations then identify their quotient spaces.

Theorem 7.3 (universal completion)**.** *For the universal presentation $\mathcal{U}_X$, scoped rewrite equality of parallel traces is equivalent to endpoint-fixed homotopy of their realizations. Consequently,*

$$G_{\mathcal{U}_X} \cong \Pi_1^q(X)$$

as abstract groupoids, and the comparison is a homeomorphism on arrow spaces.

Proof. Soundness is the general soundness theorem. Conversely, an endpoint-fixed homotopy is itself one of the named universal generators, so it is a scoped rewrite in one step. The equivalence of quotient relations follows after including endpoint equality. The coherent-path quotient identification lemma supplies the missing topological identification with $\Pi_1^q(X)$, and the geometric completeness theorem then gives the homeomorphism and groupoid isomorphism. □

Corollary 7.4 (collapse of the two universal topologies)**.** *For the universal presentation, the comparison map from Proposition 2.4*

$$J_{\mathcal{U}_X} : G^{\mathrm{tr}}_{\mathcal{U}_X} \longrightarrow G^{\mathrm{obs}}_{\mathcal{U}_X}$$

is a homeomorphism. More precisely, both spaces are canonically homeomorphic to $\Pi_1^q(X)$, and these homeomorphisms identify $J_{\mathcal{U}_X}$ with the identity on path classes.

Proof. The coherent-path quotient identification lemma applies to both representative topologies and identifies each universal scoped quotient with $\Pi_1^q(X)$. The quotient maps are induced by the same representative realization γ, so the two identifications commute with the identity on path classes. Hence the comparison map in Proposition 2.4 is the composite of one homeomorphism with the inverse of the other, and is therefore a homeomorphism. □

Proposition 7.5 (the universal path projection)**.** *Let $q_I^{\mathcal{U}} : X^I \to G^{\mathrm{obs}}_{\mathcal{U}_X}$ send a path to the class of its one-letter coherent representative. This map is quotient.*

If $q_I^{\mathcal{U}}$ is open, the universal presentation has product-quotient compatibility, and multiplication is continuous on the ordinary pullback of its arrow space.

Proof. The geometric-path projection $r : T^{\mathrm{obs}}_{\mathcal{U}_X} \to X^I$ is continuous and has the continuous one-letter section σ. Every coherent representative has the same scoped class as the one-letter representative of its geometric path. Thus $q_T = q_I^{\mathcal{U}} \circ r$ and $q_I^{\mathcal{U}} = q_T \circ \sigma$. The first identity and the quotient property of q_T make $q_I^{\mathcal{U}}$ quotient.

On composable pairs, $r \times r$ again has a continuous section, so its restriction to the endpoint pullbacks is quotient. If $q_I^{\mathcal{U}}$ is open, its square remains an open quotient after restriction to the full inverse image of composable arrow pairs. The composite is the ordinary pair map for $\mathcal{U}_X$. The ordinary/final comparison theorem now gives product-quotient compatibility and continuity of multiplication. □

Theorem 7.6 (a positive geometric class)**.** *Let X be locally path-connected and semilocally simply connected. The universal observable presentation $\mathcal{U}_X$ has product-quotient compatibility. Its scoped quotient is therefore a topological groupoid for the ordinary pullback topology. For every $x \in X$, the based arrow fiber $G_{\mathcal{U}_X}(x, x)$ with the subspace topology from the global arrow space is discrete, and its multiplication is continuous on the ordinary product.*

Proof. Holkar, Hossain, and Kulkarni prove that the path-class projection $q_I : X^I \to \Pi_1^q(X)$ is open under these hypotheses [10, Corollary 3.7]; their Theorem 3.9 also identifies the resulting ordinary topological groupoid. The companion Lean proof makes the openness argument explicit. Let W be open in X^I, and choose $\gamma \in W$. For sufficiently small $\varepsilon > 0$, the path

$$t \longmapsto \gamma\big(\max(0, \min(1, (1 + 2\varepsilon)t - \varepsilon))\big)$$

still belongs to W. It is constant near both endpoints. Choose a finite compact-open subbasic neighborhood of this reparametrized path inside W. Intersecting the target sets that contain $\gamma(0)$ gives an open set U around $\gamma(0)$; do the same at $\gamma(1)$ to obtain V. Any paths $\alpha : a' \to \gamma(0)$ and $\beta : \gamma(1) \to b'$ whose ranges lie in U and V can be inserted into the two constant tails. The resulting path remains in W and is homotopic relative to its endpoints to $\alpha\gamma\beta$. The latter claim follows by a piecewise linear change of speed.

Now suppose δ represents the same path class as γ. The finite null-homotopy ladder, using local path-connectedness and semilocal simple connectivity, gives a compact-open neighborhood N of δ. Each $\zeta \in N$ is homotopic to $\alpha\delta\beta$ for endpoint connectors with ranges in U and V. Replacing δ by γ preserves that homotopy class, and the absorption construction gives a representative in W. Thus N lies in the saturation of W. The saturation is open, so the quotient map q_I is open.

The arrow-space homeomorphism of Theorem 7.3 identifies q_I with $q_I^{\mathcal{U}}$. Proposition 7.5 therefore gives product-quotient compatibility and continuous ordinary multiplication.

The based arrow fiber is carried by that homeomorphism to the subspace of $\Pi_1^q(X)$ with both endpoints x. This subspace has the usual based-loop quotient topology: the based-loop space is the full inverse image of the fiber under the open map q_I, so the restricted map is still open and quotient. The one-letter section gives the same quotient topology from the coherent based fiber. Local path-connectedness and semilocal simple connectivity make the quotient fundamental group discrete [10, Corollary 3.10(iii)]. Its product is therefore discrete, and multiplication is continuous. □

Corollary 7.7 (totally disconnected spaces)**.** *If X is totally disconnected, $q_I^{\mathcal{U}}$ is a homeomorphism. The universal presentation has product-quotient compatibility and continuous multiplication on the ordinary composable-pair space. This conclusion does not require local path-connectedness.*

Proof. The interval is connected, so every continuous path in X is constant. Two such paths with the same scoped class have the same source and are therefore equal. Thus $q_I^{\mathcal{U}}$ is injective. It is quotient by Proposition 7.5, hence a homeomorphism. The remaining claims follow from the resulting openness of $q_I^{\mathcal{U}}$ and the open-projection criterion. □

The Hawaiian earring in Proposition 4.9 is locally path-connected and fails the semilocal hypothesis at its common base point. These examples locate the ordinary-pullback obstruction in geometric presentations. The groupoid theorem applies the published compact-open quotient result; our construction supplies the presentation-level comparison.

8. Finite presentations and their products

These examples use traces whose endpoints are the chosen base point. Throughout this section, $G_{\mathcal{P}}(a,a)$ means the quotient of the coherent based-loop carrier $T(a,a)$ with its own quotient topology. It is not assumed to have the subspace topology inherited from the global arrow quotient. We first reduce words by the scoped rules, then use winding numbers to distinguish the resulting normal forms.

8.1. **The additive circle: a finite generator and its completion.** Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ carry the quotient topology. The minimal presentation has one oriented primitive step a at the base point 0, with realization

$$a(t) = [t] \in \mathbb{T}.$$

The formal reversal supplies a^{-1}; no integer-indexed primitive step is assumed. A based trace is therefore a finite signed word in a and a^{-1}. No additional cancellation generator is needed: the structural inverse coherences in Definition 2.12 instantiate to $a; a^{-1} \simeq_{\mathcal{P}} 1_0$ and $a^{-1}; a \simeq_{\mathcal{P}} 1_0$, and are closed under the scoped congruence.

For a signed word w, let $\exp(w)$ be the number of positive letters minus the number of negative letters. Let c_n be the canonical word consisting of n copies of a when $n \geq 0$ and $-n$ copies of a^{-1} when $n < 0$.

Lemma 8.1 (finite-generator reduction)**.** *Every signed word w has a finite scoped derivation*

$$w \simeq_{\mathcal{P}} c_{\exp(w)}.$$

If two canonical words have endpoint-fixed homotopic realizations, then their exponents are equal. In particular, scoped-equivalent canonical words have equal exponents by soundness.

Proof. Scan the word from left to right while retaining a reduced signed word. If the next sign agrees with the last retained sign, append it. If the signs differ, delete the adjacent inverse pair using the corresponding structural coherence inside the surrounding prefix and suffix, using concatenation congruence. Each deletion strictly decreases the word length, so the procedure terminates. A terminal reduced word has no adjacent sign change and hence consists entirely of positive letters or entirely of negative letters. The integer exp is unchanged by every deletion, so the

terminal word is exactly $c_{\exp(w)}$. The same invariant is the endpoint of the lift to $\mathbb{R}$; therefore endpoint-fixed homotopy of canonical words forces equality of their exponents. □

For example, the cancellation mechanism gives the concrete scoped reduction

$$a; a^{-1}; a^{-1}; a; a \ \simeq_{\mathcal{P}} \ a^{-1}; a; a \ \simeq_{\mathcal{P}} \ a.$$

The first step cancels the initial $a; a^{-1}$ pair and the second cancels the remaining $a^{-1}; a$ pair.

The reduction lemma is the normal-form certificate from Section 6: the reduction derivation gives clause (a), and the lifted endpoint gives clause (b). Thus the minimal circle presentation is geometrically complete on its based loop fiber.

The derived integer normal forms also give a completed presentation with a primitive step λ_n for each $n \in \mathbb{Z}$,

$$\lambda_n(t) = [tn].$$

Its named rules are the derived integer rules

$$(n; m) \simeq_{\mathcal{P}} (n + m), \qquad 1_0 \simeq_{\mathcal{P}} 0, \qquad (n)^{-1} \simeq_{\mathcal{P}} (-n).$$

This integer-indexed presentation packages normal forms already derived from the single generator. The proof of the finite-generator classification does not use it.

Let L be the endpoint-fixed homotopy quotient of the based circle loop space, with its quotient topology. The map sending a loop class to its winding number and the map sending n to the standard loop class are inverse maps. Under the completeness conclusion below, write

$$S_0 : L \longrightarrow G_{\mathcal{P}}(0, 0)$$

for the inverse of the realization comparison on the based-loop fiber.

Theorem 8.2 (circle winding classification)**.** *The finite-generator scoped presentation is geometrically complete on the based loop fiber, and winding number is a group isomorphism and a homeomorphism*

$$L \cong \mathbb{Z},$$

where $\mathbb{Z}$ is discrete. Moreover, the inverse realization comparison

$$S_0 : L \xrightarrow{\cong} G_{\mathcal{P}}(0, 0)$$

is a homeomorphism. In particular the scoped presentation is nondegenerate: the images of the zero and one winding loops are distinct.

Proof. Lift a based loop to $\mathbb{R}$ starting at zero. Its integer endpoint is homotopy invariant; straight-line interpolation gives the standard representative. Scoped cancellation yields the same normal form. Uniformly close circle loops have homotopic lifts, so winding is locally constant. The discrete normal-form section gives the inverse homeomorphism. The variable-base winding map supplies the separate global-fiber comparison. Detailed calculations are in the Supplement, Section S3. □

8.2. **The torus and its two generators.** Let

$$\mathbb{T}^2 = (\mathbb{R}/\mathbb{Z}) \times (\mathbb{R}/\mathbb{Z})$$

with base point $(0,0)$. Use two finite oriented generators

$$a(t) = ([t],[0]), \qquad b(t) = ([0],[t]).$$

The only named rule is the commuting square

$$a; b \simeq_{\mathcal{P}} b; a;$$

inverse cancellations for a and b are structural, as for the circle. The commuting square is sound: it is the projection modulo $\mathbb{Z}^2$ of the evident square homotopy in $\mathbb{R}^2$. The other signed commutations are derived, rather than added as generators. For example, with unit insertions and structural cancellations displayed schematically,

$$\begin{aligned} a^{-1}; b &\simeq_{\mathcal{P}} a^{-1}; (b; a); a^{-1} \simeq_{\mathcal{P}} a^{-1}; (a; b); a^{-1} \simeq_{\mathcal{P}} b; a^{-1}, \\ a; b^{-1} &\simeq_{\mathcal{P}} b^{-1}; (b; a); b^{-1} \simeq_{\mathcal{P}} b^{-1}; (a; b); b^{-1} \simeq_{\mathcal{P}} b^{-1}; a, \\ b^{-1}; a^{-1} &\simeq_{\mathcal{P}} a^{-1}; b^{-1} \end{aligned}$$

and symmetry gives the reversed orientations. Thus closure under the scoped congruence supplies all four signed commuting rules.

For a signed word w, let $m(w)$ and $n(w)$ be its signed counts of a and b. Move every b-letter to the right using the commuting square and then perform the two cancellation algorithms from the circle. This is a finite terminating derivation

$$w \simeq_{\mathcal{P}} a^{m(w)}; b^{n(w)}.$$

The pair $(m(w), n(w))$ is invariant under every scoped rule. Each signed commutation strictly decreases the number of pairs in which a $b^{\pm 1}$-letter occurs to the left of an $a^{\pm 1}$-letter. For a short worked instance, sorting and cancellation give

$$b; a; b^{-1}; a; b \;\simeq_{\mathcal{P}}\; a; b; b^{-1}; a; b \;\simeq_{\mathcal{P}}\; a; a; b \;=\; a^2; b.$$

The first step uses the commuting square and the second cancels $b; b^{-1}$; the winding pair is $(2,1)$.

Lemma 8.3 (simultaneous and sequential torus representatives)**.** *Let*

$$d_{m,n}(t) = ([mt],[nt])$$

be the simultaneous standard loop, and let $|a^m; b^n|$ be the equal-slot realization of the sequential word. Then

$$d_{m,n} \simeq_{\partial I} |a^m; b^n|.$$

Proof. Choose lifts of both loops to paths in $\mathbb{R}^2$ beginning at $(0,0)$. The lift of $d_{m,n}$ is $t \mapsto (mt, nt)$, while the lift of the sequential equal-slot realization is piecewise linear. Both lifts end at (m,n). The straight-line interpolation between these two lifted paths is therefore fixed at both endpoints. Composing it with the covering map $\mathbb{R}^2 \to \mathbb{T}^2$ gives the required endpoint-fixed homotopy. □

Let $L_{\mathbb{T}^2}$ be the endpoint-fixed homotopy quotient of the based loop space of $\mathbb{T}^2$, with its quotient topology. Under geometric completeness, write

$$S_{\mathbb{T}^2} : L_{\mathbb{T}^2} \longrightarrow G_{\mathcal{P}}((0,0),(0,0))$$

for the inverse of the realization comparison on the based-loop fiber.

Theorem 8.4 (torus winding classification)**.** *For the torus $\mathbb{T}^2$, the coordinate winding pair is a group isomorphism and a homeomorphism*

$$L_{\mathbb{T}^2} \cong \mathbb{Z} \times \mathbb{Z},$$

where $\mathbb{Z} \times \mathbb{Z}$ is discrete. The finite-generator scoped presentation is geometrically complete on the based loop fiber. Moreover, the inverse realization comparison

$$S_{\mathbb{T}^2} : L_{\mathbb{T}^2} \xrightarrow{\cong} G_{\mathcal{P}}((0,0),(0,0))$$

is a homeomorphism.

Proof. Apply circle lifting in each coordinate. The commuting-square rule sorts the two generators, and cancellation gives the integer-pair normal form. Coordinate winding is locally constant and the discrete normal-form section is continuous. The variable-base winding pair gives the separate global-fiber comparison. Detailed calculations are in the Supplement, Section S3. □

Figure 10 illustrates the lifted paths used in both proofs.

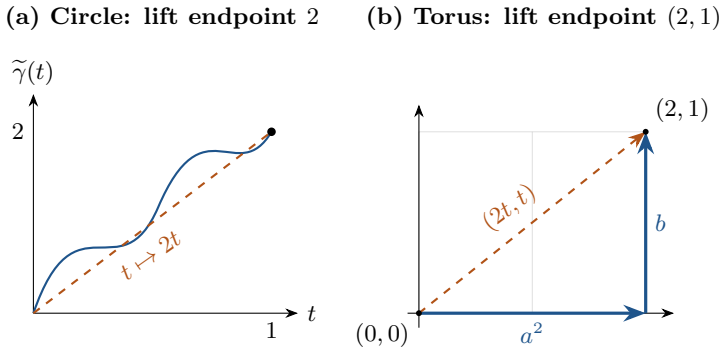


FIGURE 10. Winding separates normal forms because lifts remember endpoints. Left: a lift beginning at 0 and ending at 2 is straightened to $2t$. Right: the sequential word $a^2; b$ and the simultaneous loop have lifts with common endpoints $(0,0)$ and $(2,1)$; straight-line interpolation in $\mathbb{R}^2$ gives a homotopy after projection to the torus. Both plots are drawn in the covering spaces $\mathbb{R}$ and $\mathbb{R}^2$.

Alt text: A lifted circle loop ends at an integer; coordinate lifts of a torus loop end at an integer pair, determining their normal forms.

8.3. **Products of complete based presentations.** The torus sorting argument extends to a product without requiring normal forms for either factor. Let $\mathcal{P}$ be a sound scoped presentation on X with primitive steps E, and let $\mathcal{Q}$ be one on Y with primitive steps F. A product trace has horizontal letters (e, y') and vertical letters (x', f), where $e \in E$, $f \in F$, $x' \in X$, and $y' \in Y$. Their realizations hold the other coordinate fixed. Lift each named rule of $\mathcal{P}$ at every y', and each rule of $\mathcal{Q}$ at every x'. Add interchange around every primitive rectangle:

$$(e, y_0); (x_1, f) \simeq_{\mathcal{P} \boxtimes \mathcal{Q}} (x_0, f); (e, y_1), \quad e : x_0 \to x_1, \quad f : y_0 \to y_1.$$

The corresponding interchange for arbitrary factor traces $p : x_0 \to x_1$ and $q : y_0 \to y_1$ follows from these primitive squares:

$$H_{y_0}(p); V_{x_1}(q) \ \simeq_{\mathcal{P}\boxtimes\mathcal{Q}} \ V_{x_0}(q); H_{y_1}(p).$$

Here H_y and V_x lift entire traces while holding the other coordinate fixed. To derive the displayed relation, fix a vertical primitive step f and induct on p. The single-step case is the named rectangle rule; the unit case uses the groupoid unit laws. For a concatenation, paste the two previously derived squares along their common edge. For an inverse, reverse a square and cancel the resulting inverse pairs. Induction on q then gives the general trace relation by the same pasting and inverse steps. Write $\mathcal{P} \boxtimes \mathcal{Q}$ for the presentation with primitive rectangle rules. Its rules are sound: the two paths are the two edge routes from (x_0, y_0) to (x_1, y_1) in the square $(u, v) \mapsto (\rho(e)(u), \rho(f)(v))$. The routes are homotopic relative to their endpoints inside the square.

Theorem 8.5 (product closure of based completeness)**.** *Suppose $\mathcal{P}$ is geometrically complete on the based-loop fiber at $x \in X$ and $\mathcal{Q}$ is geometrically complete on the based-loop fiber at $y \in Y$. Then $\mathcal{P} \boxtimes \mathcal{Q}$ is geometrically complete on the based-loop fiber at (x, y).*

Proof. Primitive rectangle swaps move horizontal letters before vertical letters, including all signs. Induction sorts every product trace into factor traces. Completeness of the factors then identifies exactly the traces with homotopic geometric realizations; soundness gives the converse. Detailed calculations are in the Supplement, Section S3. □

Iterating the theorem from the finite oriented circle presentation gives completeness of the corresponding product presentation of every n-torus on its based-loop fiber. The product construction includes a horizontal or vertical copy of each factor step at every point of the other factor, so its step space is generally infinite. It is distinct from the finite-generator torus presentation of Theorem 8.4. The Lean development checks sorting and based trace completeness for a presentation that names whole-trace interchange. It also derives every such rule from primitive squares, proves equivalence of the two scoped rewrite relations, and transfers completeness to the primitive-only presentation.

8.4. **A geometric nonabelian example: the two-circle rose.** Figure 11 shows how the covering retains word order.

Let R be the quotient of two copies of $\mathbb{R}/\mathbb{Z}$ identifying their zero points, with basepoint o. It is homeomorphic to the two coordinate circles inside $(\mathbb{R}/\mathbb{Z})^2$. Let a and b be their standard once-around based loops. Give the two primitive labels the discrete topology and declare no named rewrites. Thus cancellation and the structural groupoid rules are the entire scoped presentation.

Write $F(a, b)$ for the free group on the two labels. Reading a signed trace as a group word gives $\mathrm{ev}(p) \in F(a, b)$. Its reduced word is an actual normal form: structural scoped rewriting derives p equivalent to the trace reconstructed from that word. This algebraic normalization works for any two based primitive loops. The following geometric bridge uses the particular space R.

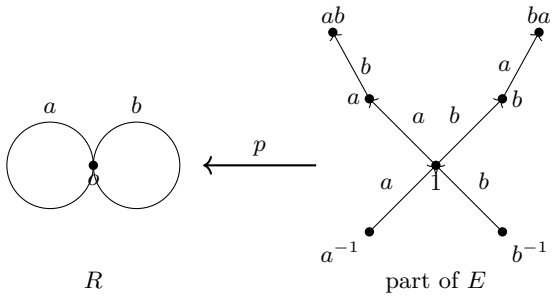


FIGURE 11. The geometric covering records word order in lifted endpoints. The two words ab and ba end at different vertices above the same basepoint. The pictured finite part belongs to the full interval-glued Cayley realization.

Alt text: The two-circle rose is covered by an interval-glued Cayley graph. Lifts of ab and ba end at distinct vertices above its basepoint.

Theorem 8.6 (represented-loop completeness for the rose)**.** *For based traces p, q in this presentation,*

$$p \simeq_{\mathcal{P}} q \quad \Longleftrightarrow \quad \mathrm{ev}(p) = \mathrm{ev}(q) \quad \Longleftrightarrow \quad |p| \simeq_{\partial I} |q|.$$

Both the trace-sensitive and equal-slot observable fixed-endpoint based scoped quotients are discrete. In particular, $a; b$ and $b; a$ are geometrically nonhomotopic.

Proof. The normal-form argument gives the first equivalence, and soundness gives the forward geometric implication. For the converse, construct a space E with vertices $g \in F(a, b)$ and one closed interval from g to gs for each $s \in \{a, b\}$. Its topology is the final topology of these closed intervals, with precisely the specified endpoint identifications. Map each interval to the corresponding primitive circle loop and each vertex to o.

Left multiplication acts continuously on E. The map to R is a quotient map: the finite union of the two unit intervals maps continuously and surjectively from a compact space to the Hausdorff rose, and the projection forgetting the free-group coordinate is open and surjective. Descent through the interval-gluing quotient preserves this conclusion. The fibers are exactly the left-multiplication orbits. This follows from the injective open-interval parametrization of each punctured circle and the fact that the two circles meet only at o.

The action has disjoint local translates. Interior points have open neighborhoods contained in a single edge. At a vertex v, extend the word distance $\|v^{-1}g\|$ affinely along each edge. This is a continuous function d_v on E. The triangle inequality for reduced-word length gives

$$\|v^{-1}w\| \leq d_v(z) + d_w(z).$$

Consequently the open sets $d_v < 1/3$ and $d_w < 1/3$ are disjoint when $v \neq w$, and translation carries one such neighborhood to the corresponding translated vertex neighborhood. The orbit map is therefore a covering map.

A finite signed word has an explicit lift starting at the identity vertex, obtained by traversing the corresponding oriented edges. Its endpoint is the vertex given by its free-group evaluation. The equal-slot realization is homotopic to the recursively

concatenated realization, so homotopic trace realizations have lifts with the same endpoint. Vertex injectivity now gives $\mathrm{ev}(p) = \mathrm{ev}(q)$ and hence scoped equality. The two reduced words ab and ba differ.

The full-word based quotient is discrete because the primitive alphabet is discrete. For the observable quotient, the four signed primitive paths are distinct: the two circle retractions and their winding numbers distinguish them. Equal-slot restriction recovers each signed letter. Each fixed-length observation stratum is finite inside a Hausdorff path space, hence discrete, and the discrete length coordinate separates the strata. Coherent-representative elimination identifies the coherent scoped quotients homeomorphically with these trace-only quotients, so they too are discrete. □

This proof is a formalized instance of the classical graph-covering method [8, Section 1.A]. The following construction supplies the additional simple-connectivity step needed to classify every continuous based loop. Write $T = E$ for this Cayley realization.

Theorem 8.7 (all continuous loops of the rose)**.** *The geometric Cayley tree T is contractible. Every continuous based loop in R is endpoint-fixed homotopic to the equal-slot realization of a reduced finite word in $a^{\pm 1}, b^{\pm 1}$. Two loops are homotopic exactly when their assigned reduced words agree. The resulting realization comparison is a bijection from the scoped based quotient onto $\pi_1(R)$.*

Proof. Write $d(g)$ for reduced-word length and P_g for the equal-slot path from the identity vertex to g. Adjacent vertices have lengths differing by one; the shorter reduced word is an initial segment of the longer. On an edge from g to $r = ga_i$, with coordinate $t \in [0,1]$, define

$$H(s,t) = \begin{cases} P_r\left(\dfrac{(1-s)(d(g)+t)}{d(g)+1}\right), & d(g) < d(r), \\ P_g\left(\dfrac{(1-s)(d(r)+1-t)}{d(r)+1}\right), & d(r) < d(g). \end{cases}$$

At a vertex g this is $P_g(1-s)$. Exact equal-slot prefix identities prove compatibility at both endpoints, including edges incident to the identity. Each fixed closed-edge formula is jointly continuous. The closed-edge quotient remains a quotient after multiplication by the compact interval, so the formulas descend to a continuous homotopy on $I \times T$. Its time-zero map is the identity, its time-one map is the identity vertex, and it fixes that vertex throughout. Thus T is contractible and simply connected.

The concrete quotient covering now gives an equivalence between the full fundamental group and the opposite free group. Its value on a realized trace is the opposite of that trace's endpoint word, computed from its unique lift. Given any continuous loop, take the reduced word of its image under this equivalence. The realized word has the same image; injectivity proves the required homotopy. Flat and binary realizations are homotopic, so the statement holds for either timing convention. Together with Theorem 8.6, this proves bijectivity of realization on the scoped based quotient. □

Mathlib composes the second path first in its fundamental-group multiplication. Accordingly, the checked endpoint equivalence has target $F(a,b)^{\mathrm{op}}$; composing with inversion gives a group equivalence with $F(a,b)$. The direct realization comparison reverses multiplication under this convention. These algebraic declarations make no additional claim about the compact-open topology on the ordinary loop quotient.

9. Checked scope and reproducibility

The current development uses Lean 4.32.0 (kernel commit `8c9756b`) and Mathlib `81a5d257`. The direct replay manifest selects 178 actual declarations: 127 theorems and 51 definitions. The exact-source verification checked their 20,013-declaration dependency closure with NanoDa, separately from the five retained Comparator configurations. Only `propext`, `Classical.choice`, and `Quot.sound` occur as axioms. The merged source tree is `4f48603`; its proof source is identical to the verified `e1d6373` head.

Flat and binary topology declarations remain separate. The literal-word relation and stratum topology bridges, matching coherent-representative elimination, variable-length word-code invariance, conservative abbreviation failure, and actual circle timing nonhomeomorphism are checked. The complete-separation theorem assumes nondiscreteness of the ordinary based quotient explicitly. Fabel's harmonic-archipelago and Hawaiian-earring facts remain external geometric inputs; no such facts are added as axioms. The rose result classifies all continuous loops algebraically, with Mathlib's opposite multiplication convention explicit. Its topological statements concern the fixed-endpoint computational quotients.

The Supplement, Section S4 maps results to declarations, assumptions, models, and commits. It distinguishes current kernel and independent replay checks from the earlier immutable registered selection. Repository manifests and exact-source logs supply reproducible verification boundaries. Challenge-side statement placeholders are distinct from the substantive checked proofs. The Supplement supplies detailed construction and classical-example proofs.

10. Conclusion

The semantics depends on the declared rewrites, exact parametrization and composable-domain topology. Continuous primitive word codes and scoped roundtrips preserve the trace-sensitive quotient. Conservative abbreviations and binary timing collisions can change the observable topology. Literal-word correspondence connects the manuscript to the implemented flat model. The circle, torus, product and geometric rose developments prove completeness for concrete presentations. The ordinary/final criterion specifies the quotient condition that yields ordinary composition continuity.

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